UNIVERSIDADE FEDERAL DE LAVRAS

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DATA-DRIVEN STRATEGIES FOR INFERRING FROM LARGE-SCALE BIBLIOMETRIC DATASETS

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Tese de doutorado apresentada à Universidade Federal de Lavras, como parte das exigências do Programa de Pós-Graduação em Estatística e Experimentação Agropecuária, área de concentração em Estatística e Experimentação Agropecuária, para a obtenção do título de Doutor.

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RESUMO

A mobilidade internacional de pesquisadores desempenha um papel significativo no cenário globalizado, potencializando a internacionalização da ciência permitindo o acesso dos cientistas a redes de colaboração transnacionais. Portanto, embora os países de origem dos cientistas móveis possam sofrer perdas de capital humano, eles estabelecem conexões internacionais que, a longo prazo, podem trazer benefícios significativos para o avanço de suas pesquisas. Essa dinâmica pode afetar o desenvolvimento socioeconômico e inovador das nações, trazendo desafios para pesquisadores e formuladores de políticas. O objetivo geral deste trabalho é avaliar o efeito da mobilidade internacional na carreira acadêmica dos pesquisadores brasileiros da área de ciências da saúde e ciências da vida, considerando as movimentações realizadas no período de 2010 a 2016. Para isso, as etapas de organização e visualização dos dados bibliométricos da Scopus serão necessárias para encontrar padrões das trajetórias dos pesquisadores brasileiros da área da saúde, bem como entender os principais países envolvidos no movimento de migração. É válido mencionar que a complexidade e o tamanho dessa base de dados podem apresentar desafios, como vieses sistemáticos e problemas de identificação de autores. Aliado a isso, para estimar os efeitos causais da mobilidade internacional dos pesquisadores ajustou-se o modelo controle sintético generalizado, para utilizar múltiplos períodos de tratamento. Este modelo foi ajustado considerando uma amostra de pesquisadores não migrantes pareados com os pesquisadores migrantes pelo método *propensity score matching*. Ao aplicar os modelos quase-experimentais, tais como o controle sintético, buscou-se compreender as dinâmicas e impactos dessa migração na carreira dos pesquisadores brasileiros, contribuindo para uma visão mais abrangente desse fenômeno. Os resultados obtidos apresentam os Estados Unidos, Canadá e Reino Unido como principais destinos da mobilidade dos pesquisadores brasileiros das ciências da saúde e ciências da vida, sugerindo que esses movimentos ocorrem principalmente para países desenvolvidos do Norte global. Nos resultados exploratórios, também foi possível observar que as áreas da Medicina, Agricultura e Bioquímica, genética e biologia molecular dominaram entre as demais áreas em número de pesquisadores migrantes e não migrantes. Os efeitos causais dos pesquisadores móveis sugerem que a mobilidade internacional leva a mudanças nas variáveis analisadas, citações recebidas, SJR e colaboração em pesquisa. Este trabalho fornece orientação sobre o uso de controle sintético em dados bibliométricos. Além disso, o estudo oferece resultados valiosos para o desenvolvimento de políticas de internacionalização da ciência. O método que empregamos também pode ser aplicado para estudar os efeitos da mobilidade internacional em outros campos do conhecimento.

Palavras-chave: Mobilidade acadêmica. Dados bibliométricos. Organização de dados. Inferência causal. Controle sintético.

ABSTRACT

International mobility of researchers plays a significant role in a globalized context, enhancing the internationalization of science by providing scientists with access to transnational collaboration networks. Thus, while the countries of origin of mobile scientists may experience losses in human capital, they establish international connections that can yield significant benefits for the advancement of research in the long term. This dynamic can influence the socioeconomic and innovative development of nations, posing challenges for researchers and policymakers. The overarching goal of this study is to assess the impact of international mobility on the academic careers of Brazilian researchers in the fields of health and life sciences, focusing on movements between 2010 and 2016. To achieve this, it is necessary to organize and visualize bibliometric data from Scopus to identify patterns in the trajectories of Brazilian health researchers and understand the main countries involved in migration flows. It is worth noting that the complexity and scale of the database present challenges, such as systematic biases and difficulties in author identification. In addition, a generalized synthetic control model was employed to estimate the causal effects of researchers' international mobility, accounting for multiple treatment periods. This model was applied using a sample of non-migrant researchers matched to migrant researchers through the propensity score matching method. By applying quasi-experimental models like synthetic control, the study sought to understand the dynamics and impacts of this migration on the careers of Brazilian researchers, providing a broader perspective on the phenomenon. The findings highlight the United States, Canada, and the United Kingdom as the primary destinations for Brazilian researchers in health and life sciences, indicating that these movements predominantly occur toward developed nations in the Global North. Exploratory results also revealed that Medicine, Agriculture, and Biochemistry, Genetics, and Molecular Biology emerged as the dominant fields among both migrant and non-migrant researchers. The causal effects suggest that international mobility leads to changes in the variables analyzed, including citations received, SJR (Scientific Journal Rankings), and research collaboration. This study offers guidance on using synthetic control models in bibliometric data analyses. Furthermore, it provides valuable insights for the development of science internationalization policies. The methodology applied can also be extended to analyze the effects of international mobility in other fields of knowledge.

Keywords: Academic mobility. Bibliometric data. Data wrangling. Causal inference. Synthetic control.

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FIRST PART

1 INTRODUCTION

International academic mobility represents an event that must be considered in the globalized world, as it involves migratory movements of highly skilled individuals. Phenomena that have emerged within these migratory flows and to which scholars are paying increasing attention include “brain drain”, “brain gain”, and “brain circulation”. These phenomena can lead to losses or gains for the regions and countries involved, as they facilitate the transfer of human capital. Mobility plays a fundamental role in the production of scientific knowledge, enabling knowledge to move alongside mobile scientists and be utilized and shared in different locations (SCELLATO et al., 2015). Countries that host international researchers and encourage collaboration tend to produce publications with greater scientific impact (JONKERS; CRUZ-CASTRO, 2013; AYKAC, 2021).

The migratory flows of researchers can result in both gains and losses of talent for the countries involved, where developed countries tend to attract talent to the detriment of developing countries (MARGINSON, 2006; DOCQUIER et al., 2007). The concept of “brain drain” was first understood by Portes (1976) as a phenomenon involving the emigration of highly skilled personnel from certain countries, corresponding to an unexpected gain (“brain gain”) for others. However, another phenomenon frequently addressed in international mobility studies is “brain circulation,” which offers an alternative for promoting the exchange of highly qualified resources between developing and developed countries. This approach transforms migrants into links between local and global networks of science and technology (PELLEGRINO, 2001).

The emigration of highly skilled individuals tends to be stronger from developing countries to developed countries (MARGINSON, 2006). In particular, Beine et al. (2008) show that countries experiencing brain drain are primarily located in Africa and Latin America. According to UNESCO data, in 2022 there were 87,337 Brazilian higher education students abroad, with most of them located in North America and Western Europe (59.19%) (UNESCO, 2024). However, empirical findings suggest that Brazil does not suffer substantial losses of scientific talent abroad, as the predominant movement is circulation (LOMBAS, 2017).

In this context, programs and policies that promote short-term exchanges abroad and ensure the return of researchers have gained importance, as they enable collaborative ties with other countries (CAÑIBANO et al., 2011). An example of a higher education internationalization policy is the “Science without Borders” program, launched in 2011, which facilitated the temporary mobility of 92,862 Brazilian students until 2016 (ZAHLER; MENINO, 2018). Furthermore, between 2010 and 2016, the Brazilian government made significant investments in internationalization policies and the expansion of international mobility scholarships in higher education (FINARDI; GUIMARÃES, 2017; MAUÉS; BASTOS, 2017).

In the healthcare field, the emigration of qualified professionals represents a significant challenge, as strengthening and consolidating national scientific research capacity is crucial for building a knowledge base capable of addressing emerging public health challenges, such as new infectious diseases (KUEHN, 2007; EYAL; HURST, 2008). In low- and middle-income countries, scientific research has the potential to improve the health conditions of both individuals and populations, fostering the development of these nations (NASS et al., 2009; AL-WORAFI, 2024). A study on the scientific and technological contributions of Latin American and Caribbean countries during the Zika epidemic highlighted Brazil's significant participation, which was essential in combating the epidemic and in producing and disseminating knowledge about the virus and its clinical manifestations (MACHADO-SILVA et al., 2019).

Analyzing scientific mobility is a challenging task, mainly due to the lack of consistent data and precise migration indicators. Most research conducted in Brazil relies on qualitative approaches (ARCHANJO, 2017) or focuses on specific groups or programs (LOMBAS, 2017; MCMANUS; NOBRE, 2017). In recent years, various methodological approaches have been employed to study the effects of scientific mobility on a broader scale (ROBINSON-GARCIA et al., 2019; XUE et al., 2021). To estimate the effects of this phenomenon quantitatively, causal inference techniques and methods, such as quasi-experimental models, are frequently adopted (ARRIETA et al., 2017; AYKAC, 2021).

Quasi-experimental models are used to evaluate the effects of an intervention on the outcome of a variable of interest, particularly when the assignment of the experiment is not random (COOK et al., 2002). Various quasi-experimental models have been employed to estimate causal effects on the international migration of researchers, including regression discontinuity (BARUFFALDI et al., 2020; XUE et al., 2021), difference-in-differences (ARRIETA et al., 2017), matching methods (FAN et al., 2022), and synthetic control (AYKAC, 2021). Although some international studies have already used quasi-experimental methods to examine the causal effects of academic mobility, there are no records of large-scale research addressing this topic in Brazil or other Latin American countries.

In this study, statistical methods will be employed to generate evidence on the dynamics and characteristics of academic mobility among researchers in the health sciences and life sciences fields affiliated with Brazilian institutions, as well as the impact of this mobility on national scientific production. A synthetic control method approach will be used to evaluate the impacts of international migration on the careers of these Brazilian researchers. The main research questions guiding this work are: (i) What is the pattern of international mobility among researchers in the health sciences and life sciences fields affiliated with Brazilian institutions? (ii) Which thematic areas are associated with higher scientific mobility? (iii) What is the impact of researcher mobility on national scientific production?

The publications of Brazilian researchers in the study's focus areas are extracted from the Scopus database, providing information such as the authors' country of affiliation, the number of citations per article, and collaborations. To assess article quality, we use the SJR (Scimago Journal Rank) index. Based on the authors' affiliations, researchers are classified into different types of academic mobility. The final dataset covers Brazilian publications between 2000 and 2020, categorized into international mobility types (non-migrants, immigrants, emigrants, return migrants, and transients). A careful preprocessing step is conducted to mitigate potential biases inherent to the study.

This work is being developed with data from the project "Mobility networks of health researchers: brain drain or qualified return?", funded by the Carlos Chagas Filho Research Support Foundation of the State of Rio de Janeiro (FAPERJ) and coordinated by Bruna de Paula Fonseca e Fonseca (Fundação Oswaldo Cruz - Fiocruz).

The doctoral thesis is structured as follows: the first part presents the Objectives (2), the foundation of the Literature Review (3), the steps developed with the data in Data Pipeline for Academic Mobility Analysis (4), and this Introduction (1). The second part introduces the two articles generated from the analysis of the thesis problem and the software package developed in *R*. Finally, the third part contains the Final Considerations (5).

In the first article, we describe the process of collecting, organizing, and analyzing data from Scopus, aiming to present the details involved in each step of the bibliometric study to serve as support material for future studies in different fields of knowledge. In this initial study, we adopted a data science-oriented approach to conduct qualitative analyses of the academic mobility of Brazilian researchers in the health and life sciences fields. Using data science methods and descriptive statistical techniques, we sought to explore mobility patterns, identify trends, and generate detailed insights into mobility dynamics. The application of data science allows for the precise analysis of large volumes of data, enriching the understanding of academic mobility and providing a solid foundation for future investigations.

In the second article, we conducted an in-depth causal analysis to evaluate the impact of academic mobility on scientific productivity, leveraging the data organized in the first study. We proposed the use of the generalized synthetic control method, a robust technique for estimating causal effects in scenarios with multiple treatment periods (represented here by the departure of researchers from Brazil). To enhance the precision of the estimates, we applied a matching method to reduce the sample of potential control cases, strengthening the comparative base and ensuring a more rigorous analysis of the effects of mobility on scientific production over time.

The third scientific work included in this thesis concerns a package developed in *R*, named *capesData*, containing data on academic mobility scholarships funded by CAPES (Coordination for the Improvement of Higher Education Personnel), a Brazilian federal government funding agency. The CAPES database refers to international mobility programs for the period from 2010 to 2019. This package organized and processed the data into a single dataset that can be easily analyzed using *R*.

In the third part, we synthesize the main findings of the research, discussing their contributions to understanding academic mobility in Brazil. We highlight the most relevant findings that demonstrate the effects of mobility on scientific production and the development of international collaboration networks. Additionally, we address the limitations of the research, such as the reliance on bibliometric data, which may not fully capture the complexity of academic mobility, and the possible constraints in accessing updated and complete data. Finally, we suggest directions for future studies, such as exploring other methodological approaches, including longitudinal analyses with data from different sources, or examining additional variables, such as the impact of mobility on specific research areas and the development of new indicators to more thoroughly track the dynamics of academic mobility.

2 OBJECTIVES

2.1 General Objectives

The general objective of this work is to evaluate the effects of international migration of Brazilian researchers in the health field on scientific production, considering variables related to recognition, “quality”, and research collaboration. The aim is to understand academic mobility patterns and the differences among the types of mobility investigated. The main question to be answered is: “What are the differences in the scientific production of Brazilian health researchers after emigrating from their country of origin?”

2.2 Specific Objectives

The specific objectives of this work include:

i Perform data preprocessing and organization, and define groups of authors according to international mobility, namely: immigrant, emigrant, non-migrant, return migrant, and transient;

ii Conduct an exploratory analysis of Scopus data using descriptive statistics and data visualizations to understand the trajectories of mobile Brazilian researchers in the health field;

iii Evaluate the effects of international mobility of researchers on scientific production by fitting quasi-experimental models to data on scientific indicators, such as the number of citations and SJR index;

iv Develop a package with data from the CAPES agency on academic mobility scholarships for Brazilian researchers.

3 LITERATURE REVIEW

This literature section provides a comprehensive overview of the methodologies used to analyze the impact of academic mobility, exploring causal inference approaches, quasi-experimental models, and the synthetic control method, as well as their applications in different contexts and variations of the model.

3.1 Causal Inference

Studies in the area of causal inference began in the mid-1920s with Wright (1921) and Neyman (1923), whose objective was to define causal effects in randomized experiments and extensions to observational studies. In the following decades, the field of causal inference was again studied by Wold (1956) in order to examine the differences in exploratory procedures when dealing with both types of data, experimental and observational. Subsequently, the field of causal inference developed from the studies of Rubin (1974), Rosenbaum (1984) Holland (1986), Pearl (2010) and Imbens e Rubin (2015).

Causal inference seeks to understand the cause and effect relationships of various problems in the contemporary world. In other words, inference aims to evaluate the effect of a certain cause on a certain result. According to Pearl (2010), causal inference aims to infer the dynamics of events based on changing conditions, such as treatments, interventions or policies. Several methods have already been used to make causal inferences on topics from various areas of knowledge, such as social and economic impacts caused in China during the COVID-19 pandemic (QIU et al., 2020), effects of public policies on human health (CHEN et al., 2018), causal effects of school closures in reducing the spread of COVID-19 (FUKUMOTO et al., 2021) and effects of public policies on agriculture (ITO et al., 2019). Therefore, causal inference works in several scenarios to estimate the causal effects of a treatment on a given response variable.

Rubin (1974) points out that non-random experiments can estimate useful causal effects, just like randomized experiments. Although the choice for randomized experiments should be made whenever possible, in several situations it is not viable, such as social science events. Thus, an alternative to the problem is to use estimates of the causal effects of the treatment in the non-random event, through a quasi-experimental model. Furthermore, there are studies in which the experiments cannot be randomized, whether due to the cost involved in randomization, ethical issues or too much time to obtain estimates.

The estimation of causal effects occurs with the identification of a certain event, which exerts a change in the variable of interest, when compared to a control group. The central problem in identifying a causal effect is that the variable of interest is observed under either treatment or control, but not under both factors (DEHEJIA; WAHBA, 2002).

Considering that W is the treatment and Y the result, formally, Y^W refers to the potential result in the presence and absence of treatment. The binary variable W is responsible for indicating the units that participated in the treatment ($W = 1$) or did not participate in the treatment ($W = 0$). Therefore, Y^1 is the value of the variable of interest when the unit is subjected to the treatment, and Y^0 is the value of the variable of interest when the unit is not subjected to the treatment. According to Heinrich et al. (2010), the observed result can be denoted as:

$$Y = (1-W)Y^0 + WY^1. \tag{3.1}$$

Therefore, based on the potential results, the causal effects of the treatments can be estimated. According to Rosenbaum e Rubin (1983), the average treatment effect (ATE) is given by

$$ATE = E(Y^1) - E(Y^0), \tag{3.2}$$

where $E(.)$ indicates the expectation of the population of interest. The Equation 3.2 is the difference between the effect of the treatment for the treated population and the effect of the treatment on the untreated population, therefore, it is not possible to know $E(Y^0)$ and $E(Y^1)$ in the same unit.

Heinrich et al. (2010) also highlight the average effect of the treatment on the treated (*Average Treatment Effect on the Treated - ATT*), which measures the effect of the treatment on the units that participated in the treatment, that is,

$$ATT = E(Y^1 - Y^0 | W = 1). \tag{3.3}$$

In this case, through the treated population, there is the scenario in which this population received the treatment ($W = 1$) and did not receive it ($W = 0$), but, again, it is not possible to obtain data for the situation of what would have happened without the treatment in the same population, that is, the counterfactual. Faced with such a situation, a question to ask is how to use observational data to link observed results to potential outcomes? Data from observational (quasi-experimental) studies can be used to provide estimates of what the response variable would be like in the absence of treatment. This can be done differently between quasi-experimental models, as well as whether a control group is used or not.

3.2 Quasi-experimental models

Quasi-experimental models seek to estimate the effects of an intervention policy on the result of a variable of interest, for events that do not consider the random assignment of treatments (COOK et al., 2002). This suggests that this assignment happens through self-selection, with units choosing the treatment for themselves. The term "quasi-experimental" is denoted by Rosenbaum (1984), Holland (1986) and Abadie e Gardeazabal (2003) as "observational studies", but although there is a difference in the term, they continue to consider that the attribution treatment in these cases did not occur randomly.

Meyer (1995) calls these studies "natural experiments", the purpose of which is to examine outcome measures for observations in treatment groups and comparison groups that are not randomly assigned. Furthermore, the author points out that the term "quasi-experiments" emphasizes that these studies are not exactly experiments, unlike the term "natural experiments", which suggests that these studies are experiments and occur spontaneously. Thus, Meyer (1995) concludes that the term "natural experiments", which is more commonly used in economics, somewhat inappropriately suggests non-randomized studies.

However, quasi-experimental designs generally create less convincing support for counterfactual inferences, given the need to create a treaty-like group to make (COOK et al., 2002) comparisons. In this sense, Meyer (1995) highlighted the threats that non-random experiments can suffer for internal validity, such as, for example, biases from omitted variables, trends in the result, measurement errors and simultaneity of treatments. Therefore, in quasi-experimental studies, the limitations of each model must be considered and the comparison between the treated and control groups must be carried out carefully.

Methods used to study quasi-experimental events include regression discontinuity, interrupted time series, and differences-in-differences (DINARDO, 2010). These methods have the common objective of investigating the effects of an intervention on the response variable, but present differences in the statistical models used and the use of comparison groups.

3.3 Synthetic Control

The synthetic control method aims to analyze the effects of an intervention on a variable of interest for a specific unit (country, region, municipality, etc.) in the period following the intervention. This method for estimating causal effects was proposed by Abadie e Gardeazabal (2003), where the synthetic trajectory of the variable of interest for treated regions over time is constructed based on multiple control units. Various applications in observational studies use synthetic control as an alternative to estimate causal effects, whether in public policy evaluations for combating smoking (ABADIE et al., 2010), analyses of vaccination incentive policies (SEHGAL, 2021), impacts of natural disasters on electoral outcomes (MASIERO; SANTAROSSA, 2021), or analyses of the economic effects of public policies (MATTI, 2024).

Evaluating the effects of specific interventions and education policies is also an important task that has employed the synthetic control method to estimate intervention effects (DEMIRCI, 2023; ÁLVAREZ-MUNOZ et al., 2024; LI; LIU, 2024). In the study by Demirci (2023), synthetic control is used to investigate how the domestic political environment influences the decision of young people to study abroad and potentially emigrate. This technique highlighted changes in students' migratory behavior by comparing with a scenario without populist policies, showing how transformations in the political environment can indirectly affect educational and academic mobility. Another noteworthy example is the work of Aykac (2021), which seeks to analyze the effects of researchers leaving Turkey on scientific production using synthetic control.

In addition to the various applications of synthetic control in different fields, Abadie (2021) explores the requirements, generalizations, and methodological aspects of this approach in his work. He highlights the importance of high-quality data and the careful selection of control units to ensure the feasibility of the models, as well as discussing the limitations and potential adaptations of the method to handle multiple treatment periods and variations in synthetic weights. These guidelines expand the applicability of the method and the understanding of its effectiveness in robust comparative studies.

The classic synthetic control approach uses a weighted average of unaffected regions to construct the counterfactual, which is then compared to the single treated unit (ABADIE; GARDEAZABAL, 2003). However, a generalized version of this model was developed by Xu (2017), capable of working with multiple treatment units and across different treatment periods. Using a generalized synthetic control methodology, Thomas et al. (2024) investigated the effects of a Brazilian public policy on hospitalizations and mortality among healthcare system users.

Additionally, Abadie et al. (2010) highlight that synthetic control can generate informative inferences, even with few comparison units and different levels of aggregation. According to Abadie (2021), this approach is based on the idea that a weighted combination of units in the donor pool can better replicate the characteristics of the treated unit than any single unit. Matching methods, such as propensity score matching, are often used to select more similar control units, further enhancing the comparative validity of the results.

3.3.1 Model settings

In the classical formulation of synthetic control for a treated unit, $J+1$ units are considered, where the first unit is exposed to the intervention, and the remaining J units serve as controls, forming the donor pool (ABADIE et al., 2010). Let Y_{it}^{N} be the value of the response variable without intervention for units $i = 1, \ldots, J+1$ over the periods. With T_0 indicating the number of pre-intervention periods ($1 \leq T_0 < T$), we define Y_{it}^{I} as the observed outcome for unit i in period t if unit i is treated after T_0. The intervention effect of interest for the affected unit in period t (with $t > T_0$) is

$$\alpha_{1t} = Y_{1t}^{I} - Y_{1t}^{N} \quad . \tag{3.4}$$

The main challenge in policy evaluation is estimating Y_{1t}^{N} for $t > T_0$, that is, predicting how the outcome of the treated unit would have evolved in the absence of the intervention — a counterfactual outcome, because the unit was exposed to the intervention after $t = T_0$. As Equation 3.4 shows, since Y_{1t}^{I} is observed, the estimate of the intervention impact boils down to inferring Y_{1t}^{N}. The equation also allows for the intervention effect to vary over time, acknowledging that the impacts may be gradual, cumulative, or dissipative as the post-intervention periods unfold (ABADIE et al., 2010).

According to Abadie (2021), comparative case studies aim to reproduce the value of the response variable that would have been observed for the affected unit in the absence of the intervention, based on an unaffected unit or a small number of unaffected units with similar characteristics to the affected unit at the time of intervention. Formally, a synthetic control can be represented by a $J \times 1$ vector of weights, $\mathbf{W} = (w_2, \ldots, w_{J+1})'$. Given a set of weights, $\mathbf{W}$, the synthetic control estimators for Y_{1t}^{N} and α_{1t} are, respectively,

$$\widehat{Y_{1t}^{N}} = \sum_{j=2}^{J+1} w_j Y_{jt} \tag{3.5}$$

$$\widehat{\alpha_{1t}} = Y_{1t} - \widehat{Y_{1t}^{N}} \tag{3.6}$$

Abadie e Gardeazabal (2003) and Abadie (2005) suggest choosing the weights w in such a way that the resulting synthetic control best resembles the pre-intervention values for the treated unit's outcome variable predictors. That is, given a set of non-negative constants, $v_1, \ldots, v_k$, they propose choosing the synthetic control, $\mathbf{W}^* = (w_2^*, \ldots, w_{J+1}^*)'$, that minimizes

$$||X_1 - X_0W|| = \left(\sum_{h=1}^{k} v_h(X_{h1} - w_2X_{h2} - ... - w_{J+1}X_{hJ+1})^2\right)^{1/2} \tag{3.7}$$

subject to the constraint that $w_2, ..., w_{J+1}$ are non-negative and sum to one. Then, the estimated treatment effect for the treated unit at time $t = T0+1, ..., T$ is

$$\widehat{\alpha_{1t}} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \quad . \tag{3.8}$$

One possibility is to choose V so that the resulting synthetic control region approximates the trajectory of the outcome variable of the affected region during the pre-intervention periods. Alternatively, the choice of V can be made such that the synthetic control $W(V)$ minimizes the mean squared prediction error (MSPE) of this synthetic control relative to Y_{1t}^N (ABADIE et al., 2010).

The synthetic control estimator, under certain conditions, is unbiased for a vector autoregressive model and provides a bias bound for a linear factor model (ABADIE et al., 2010). The linear factor model, which can be viewed as a generalization of the difference-in-differences approach, for $\widehat{Y_{1t}^N}$ is given by

$$Y_{jt}^N = \delta_t + \theta_t Z_j + \lambda_t \mu_j + \varepsilon_{jt}, \tag{3.9}$$

where δ_t is an unknown common factor with constant factor loadings across units, Z_j is a $(r \times 1)$ vector of observed covariates (unaffected by the intervention), θ_t is a $(1 \times r)$ vector of unknown parameters, λ_t is a $(1 \times F)$ vector of unobserved common factors, μ_j is a $(F \times 1)$ vector of unknown factor loadings, and ε_{jt} are the unobserved errors at the regional level with mean zero. Equation 3.9 generalizes the usual difference-in-differences model, which can be obtained if λ_t is constant for all t, meaning the model accounts for unobserved confounding factors, as long as they are constant over time.

According to Abadie et al. (2010), there is often no set of weights that exactly satisfies equation 3.7 with the observed data, so the synthetic control is chosen to make equation 3.7 approximately equal to the data. In practice, the condition $X_1 = X_0W$ is replaced by the approximate version $X_1 \approx X_0W$. However, for any specific dataset, there are no prior guarantees regarding the magnitude of $X_1 - X_0W^*$. As stated by Abadie (2021), the synthetic control displays the contributions of each unit to the counterfactual, with sparse and interpretable coefficients. Additionally, Abadie (2021) discusses dimensionality reduction, recommending restricting the control set to units with predictors closer to the treated unit in situations with multiple solutions.

3.3.2 Generalized Synthetic Control

The Generalized Synthetic Control (GSC) expands the traditional synthetic control estimator by allowing for multiple treated units and varying treatment periods, in addition to providing uncertainty measures such as standard errors and confidence intervals (XU, 2017). This approach has been widely applied in policy evaluation, with examples in public health to estimate the impact of interventions (O'NEILL et al., 2020; DENG; ZHENG, 2023), in the analysis of economic and social effects during extreme events (SHERIDAN et al., 2022), in the effects of energy policies (DONG et al., 2022), and in examining the consequences of privatizing essential services, such as sanitation (SILVA et al., 2024). The versatility of the GSC makes it valuable for cases where the intervention affects multiple units and has different treatment times, enhancing the capacity for longitudinal and comparative analysis across various public policy contexts.

According to Xu (2017), this method offers several advantages, the first of which is its use with multiple treated units and varying treatment periods. By estimating the counterfactuals in a single run, it avoids the need for individual matching for each treated unit. Additionally, the GSC generates uncertainty estimates, such as standard errors and confidence intervals, and applies cross-validation to adjust the model more accurately and automatically, using more data from the control group to increase the model's efficiency.

In the generalized model, we denote the sets of units in the treatment and control groups by τ and C, respectively, with the total number of units being the sum of treated units (N_{tr}) and control units (N_{co}), i.e., $N = N_{tr} + N_{co}$. As in the classical synthetic control approach, all units are observed over T periods, but varying treatment periods can also be considered. By assumption, Xu (2017) considers that Y_{it} is given by a linear factor model:

$$Y_{it} = \delta_{it} D_{it} + x_{it}^{'} \beta + \lambda_i^{'} f_t + \varepsilon_{it} \tag{3.10}$$

in which D_{it} is the treatment indicator, equal to 1 if the unit was exposed and 0 otherwise; δ_{it} is the heterogeneous treatment effect for unit i at time t; x_{it} is a $(k \times 1)$ vector of observed covariates, $\beta = [\beta_1, ..., \beta_k]^{'}$ is a $(k \times 1)$ vector of unknown parameters, $f_t = [f_{1t}, ..., f_{rt}]^{'}$ is an $(r \times 1)$ vector of unobserved common factors, $\lambda_i = [\lambda_{i1}, ..., \lambda_{ir}]^{'}$ is an $(r \times 1)$ vector of unobserved factor loadings, and ε_{it} represents unobserved idiosyncratic shocks for unit i at time t, with $E(\varepsilon_{it}) = 0$.

An essential assumption for this functional form is that all units share the same fixed set of factors over the observed period, implying the absence of structural breaks. With potential outcomes notation, the treatment effect for a treated unit i at time t is:

$$\delta_{ij} = Y_{it}(1) - Y_{it}(0) \tag{3.11}$$

with $i \in \tau$ and $t > T_0$, $Y_{it}(1)$ represents the outcome of unit i under treatment, and $Y_{it}(0)$ refers to the counterfactual outcome for the same unit in a scenario without the intervention. For each control unit, the data generating process takes the following form: $Y_i = x_i^{'}\beta + F\lambda_i + \varepsilon_i$ with $i \in C$. When stacked from the control units set, it becomes:

$$Y_{co} = X_{co}\beta + F\Lambda_{co} + \varepsilon_{co} \tag{3.12}$$

in which $Y_{co} = [Y_1, ..., Y_{N_{co}}]$ and $\varepsilon_{co} = [\varepsilon_1, ..., \varepsilon_{N_{co}}]$ are matrices of dimensions $(T \times N_{co})$; X_{co} is a three-dimensional matrix $(T \times N_{co} \times p)$; and $\Lambda_{co} = [\lambda_1, ..., \lambda_{N_{co}}]^{'}$ is a matrix $(N_{co} \times r)$, and the products $X_{co}\beta$ and $F\Lambda_{co}$ are matrices $(T \times N_{co})$. The number of factors is determined using a "cross-validation" algorithm that utilizes information from the control group and the treated unit during the pre-intervention period (XU, 2017).

The estimator of the average treatment effect on the treated (ATT) at time t (when $t > T_0$) is given by the difference between the actual outcome and the estimated counterfactual, that is:

$$\widehat{ATT}_{t,t>T_0} = \frac{1}{N_{tr}} \sum_{i \in T} [Y_{it}(1) - \hat{Y}_{it}(0)] \tag{3.13}$$

As highlighted by Xu (2017), the estimation of the counterfactual is carried out in three steps, involving: (i) estimating a model with interactive fixed effects using only the control group; (ii) estimating the factor loadings for each treated unit, minimizing the mean squared error of the expected outcome predicted in the pre-treatment periods; and (iii) calculating the counterfactual for the treated units based on the estimates. Since $Y_{it}(1)$ is observed for treated units in the post-treatment periods, the main goal is to construct counterfactuals for each treated unit in the post-treatment periods. The causal inference problem thus turns into a missing data prediction problem.

3.3.3 Contextual Requirements

An alternative approach to GSC would be to use a difference-in-differences model with continuous treatment (CALLAWAY et al., 2024), but GSC offers some advantages that are important in this context. The GSC method proposed by Xu (2017) estimates the average treatment effect on the treated using cross-sectional time series data when the “parallel trends” assumption is likely not valid.

The conditions in the context of the investigation under which synthetic controls are appropriate tools for policy evaluation are, in general, the same as those used in any other type of comparative case study research design. First, Abadie (2021) notes that small treatment effects are indistinguishable from other shocks in the outcome of the affected unit, especially if the outcome variable of interest is highly volatile. He also adds that the availability of a comparison group should be considered, as units adopting an intervention similar to the one adopted by the unit of interest should not be included in the donor pool because they are affected by the intervention.

Additionally, requirements such as no anticipation and no interference must be considered in synthetic control. The first refers to synthetic control estimators that may be biased if economic agents with foresight react before the policy intervention under investigation, or if certain components of the intervention are implemented before the formal realization/implementation of the intervention. The second requirement pertains to the stable unit treatment value assumption (SUTVA) of Rubin (1980), which implies that there is no interference between units. In other words, the outcomes of the units are invariant to the treatments of other units. In some cases, however, an intervention may have spillover effects on units that are not directly targeted by it. Assuming that these indirect effects do not exist is a strong restriction that is often imposed in the study design or considered in the analysis of the results.

3.4 Propensity Score Matching

The Propensity Score Matching (PSM) method is used to correct selection bias in samples by adjusting for observable differences between treatment and control groups (DEHEJIA; WAHBA, 2002). Widely used in economic impact analysis, PSM allows for comparing the effects of treatment in a specific group with the effect of no treatment in a control group. Developed by Rosenbaum e Rubin (1983), it calculates the probability of a unit receiving treatment based on observable variables, making it ideal for non-randomized experiments, such as public policies and unforeseen events.

PSM is used in studies that integrate quasi-experimental methods to create robust counterfactuals in impact analyses (GHOSH, 2021; MASIERO; SANTAROSSA, 2021). This procedure helps improve accuracy by reducing selection bias, especially in samples that exhibit significant observable differences between treatment and control groups. Formally, Rosenbaum e Rubin (1983) define the conditional probability of treatment assignment ($A = 1$), given the observed covariates X, with N units, as the following expression:

$$e(x) = P(A = 1|x) \tag{3.14}$$

which assumes

$$P(A_1, \ldots, A_n | x_1, \ldots, x_n) = \prod_{i=1}^{N} e(x_i)^{A_i} \{1 - e(x_i)\}^{1-A_i}.$$

The function $e(x)$ represents the propensity score, which is the probability of a unit receiving treatment ($A = 1$), given the observable variables x. This function, therefore, indicates the likelihood of each unit being treated, based on the characteristics already observed. As pointed out by Rosenbaum e Rubin (1983), in non-randomized studies, the function $e(x)$ (Equation 3.14) is typically unknown and needs to be estimated from the observed data using models like the logit.

According to Rosenbaum e Rubin (1983), in randomized experiments, the results between two treated groups can be easily compared because the units are considered similar. In contrast, in observational studies, this direct comparison is not possible, as treatment varies between treated and untreated units (ROSENBAUM; RUBIN, 1983). In this context, Rosenbaum e Rubin (1983) emphasize the importance of meeting certain assumptions to ensure consistent results in the PSM model, highlighting particularly the conditions of balance and strong ignorability.

3.4.1 Propensity score matching assumptions

The propensity score matching assumptions are fundamental to enable valid comparisons between the treated and control groups, especially in non-randomized experiments. According to Rosenbaum e Rubin (1983), the balancing property is applied to make comparisons more robust. The balancing score function, $b(x)$, also known as the balancing score, is defined as a function of the observable covariates X. This function is constructed so that the conditional distribution of x given $b(x)$ is the same for both the treated group ($A = 1$)

and the control group ($A = 0$). In other words, for each value of $b(x)$, the observable characteristics X are balanced between the groups, allowing for a more accurate comparison between them, i.e.,

$$x \perp A|b(x).$$

The balancing score ensures that the probabilities of each unit receiving or not receiving the treatment are similar, allowing for valid comparisons between the treatment and control groups. This condition ensures that the compared groups do not have different chances of receiving the treatment, facilitating a fair evaluation of the intervention's effect. The second essential assumption for the PSM model, defined by Rosenbaum e Rubin (1983) as strongly ignorable, is based on two principles: (i) the conditional independence assumption (CIA) and (ii) common support. The conditional independence assumption posits that the potential outcomes are independent of the treatment assignment, given the observable covariates.

According to Rosenbaum e Rubin (1983), the potential outcomes Y^0 and Y^1 are unrelated to the treatment A, conditioned on the observable covariates x in randomized experiments, that is,

$$Y^1, Y^0 \perp A|x. \tag{3.15}$$

However, this condition from Equation 3.15 is not known in observational studies. Thus, Rosenbaum e Rubin (1983) introduce the concept of strong ignorability, where treatment assignment is considered strongly ignorable given the covariates v, if

$$Y^1, Y^0 \perp A|v, \qquad 0 < P(A = 1|v) < 1, \tag{3.16}$$

for all v. In other words, when treatment assignment is strongly ignorable, given the set of covariates x, that is, when 3.16 is valid with $v = x$, treatment assignment is considered strongly ignorable.

In Rosenbaum e Rubin (1983), the common support assumption (see Equation 3.16) is also emphasized, which requires that both the treatment and control groups have the same probabilities of receiving the treatment. For this, it is essential that there is an overlapping area of propensity scores between these groups. This common support condition aims to make the groups comparable, ensuring that treated units have similar matches in the control group based on the selected covariates x, facilitating the comparison of their effects.

According to Dehejia e Wahba (2002), propensity score matching is essentially a weighting scheme that determines which weights are applied to the comparison units to estimate the treatment effect, that is,

$$\hat{\tau}|_{A=1} = \frac{1}{|N|} \sum_{i \in N} \left(Y_i - \frac{1}{|J_i|} \sum_{j \in J_i} Y_j \right), \tag{3.17}$$

where N is the treatment group, $|N|$ is the number of units in the treatment group, J_i is the set of comparison units for treatment unit i, and $|J_i|$ is the number of comparison units in J_i.

In the implementation of Propensity Score Matching, three main issues must be considered: the first is whether matching will be done with or without replacement; the second concerns how many control units will be matched with each treated unit; and the third relates to the choice of the most appropriate matching method (DEHEJIA; WAHBA, 2002).

In matching with replacement, the distance between the propensity scores of the treatment and control groups is minimized, allowing each treated unit to be matched with the closest control unit, even if that control unit has been used before (DEHEJIA; WAHBA, 2002). On the other hand, when matching without replacement, Dehejia e Wahba (2002) point out that, if there are few control units similar to the treated ones, the algorithm ends up comparing units with more distant propensity scores, which may increase bias, although it might improve the precision of the estimates.

The number of control units matched to each treated unit directly affects the quality of the matching. Using only one control unit for each treated unit ensures the smallest distance between the propensity scores of the groups, but when more than one control unit is used, the precision of the estimates may increase, although bias is likely to grow as well (DEHEJIA; WAHBA, 2002). Regarding matching methods, Heinrich et al. (2010) describes three main approaches: *Nearest Neighbor*, *Radius*, and *Kernel*.

3.4.2 Matching Algorithms

To perform matching between treated and control units, there are several algorithms available, including: the Nearest Neighbor method, the Radius method, and the Kernel method.

The Nearest Neighbor method is a simple technique that selects the control unit with the closest propensity score to the treated unit's score, creating a match based on this proximity (HEINRICH et al., 2010). This method can be applied with replacement, where a control unit can be matched multiple times, or without replacement, where each control unit is used only once for matching (HEINRICH et al., 2010).

According to Heinrich et al. (2010), the second technique, known as the Radius method, defines a maximum propensity score value to match treated and control units. The treatment impact estimator can be expressed as:

$$E(\Delta Y) = \frac{1}{N}\sum_{i=1}^{N}\left[Y_{1i} - \overline{Y}_{0j(i)}\right] \tag{3.18}$$

in which, $\overline{Y}0j(i)$ is the average outcome for all the comparison units matched with unit i, $Y1i$ is the outcome for unit i, and N is the number of treated units (HEINRICH et al., 2010).

Finally, according to Heinrich et al. (2010), the Kernel method is a non-parametric matching technique that compares the outcomes of each treated unit with a weighted average of all untreated units. In this process, the highest weight is given to the units whose propensity scores are closest to the treated unit. In other words, the technique uses the weighted average of untreated units to construct the comparison unit for each treated unit.

Another important aspect, as highlighted by Heinrich et al. (2010), is that when opting for the Kernel method, it is also necessary to choose the kernel function and the bandwidth parameter. The authors point out that there is no clear rule on which algorithm to use in each context, but a crucial issue is the trade-off between bias and efficiency.

4 DATA PIPELINE FOR ACADEMIC MOBILITY ANALYSIS

This section presents the main steps for collecting, organizing, analyzing, and visualizing bibliometric data from Scopus. Here, we will demonstrate how to extract the data via web scraping, prepare and organize the entire dataset in statistical software, as well as extract exploratory analyses from the data and share and communicate the findings with the scientific community.

4.1 Data Collection and Preparation

The data collection process was carried out using the API (Application Programming Interface) provided by Scopus for data extraction. An API is a set of rules and protocols that enables communication and interaction between different software or services. For this study, we accessed the API using scripts developed in *Python*, a high-level, versatile, and open-source programming language known for its clear and readable syntax. This makes it suitable for a wide range of applications, from web development and data analysis to automation and artificial intelligence.

Specifically, we utilized the *Pybliometrics* library, which aims to assist researchers in bibliometric studies by offering a consistent and simple interface that automates data retrieval, JSON serialization, caching, and error handling (ROSE; KITCHIN, 2019). Furthermore, this library allows researchers to expedite their analyses in reproducible ways and enables scientific evaluators to automate the retrieval of assessment data. The data collection task was complicated by the limitations imposed by Scopus. The API restricts the amount of data and the number of requests that each entity (controlled by a token) can make. Due to the large volume of data to be collected, it was necessary to divide the collection by year and area, retrieving portions of the data separately for subsequent processing. Additionally, a delay was programmed between requests to comply with the API's time constraints.

The code snippet below shows the main part of data collection using the tools from the API and the library chosen for the task. The code explicitly breaks down the data collection by area and by year, separately, generating results in different (.csv) files, but organized.

```
for a in areas:
    for y in years:
        query = "SUBJAREA({}) AND (AFFILCOUNTRY(brazil OR brasil)
```

```
AND PUBYEAR = {}) AND (LIMIT-TO(DOCTYPE, \"ar\") OR LIMIT-TO(DOCTYPE,
                                                    \"re\"))".format(a, y)

s = ScopusSearch(query)
pd.DataFrame(pd.DataFrame(s.results)).to_csv("{}-{}.csv".format(a,y))
print(a + ", " + str(y) + ", " + str(s.get_results_size()))
time.sleep(120)
```

The publications extracted from Scopus correspond to the following areas within the domain of health sciences and life sciences: Agricultural and Biological Sciences; Biochemistry, Genetics and Molecular Biology; Dentistry; Health Professions; Immunology and Microbiology; Medicine; Neuroscience; Nursing; Pharmacology, Toxicology, and Pharmaceutics; Veterinary; and Multidisciplinary.

4.2 Data Wrangling and Analysis

The raw data is divided into several files, with each file containing all publications for a given author (author ID). The total number of files to be read and organized exceeds 400,000 CSV files, requiring the use of the *arrow* (RICHARDSON et al., 2024), *readr* (WICKHAM et al., 2024), and *purrr* (WICKHAM; HENRY, 2023) packages. After reading and concatenating the data into a single dataframe, it is necessary to classify authors into mobility groups, as described in (SUBBOTIN; AREF, 2021).

The preprocessing and data organization stages were subdivided into the following steps: (1) Reading CSV files; (2) Data reshaping; (3) Classifying mobility groups; (4) Adding academic journal metrics; (5) Cleaning data; and (6) Creating variables. For the first step (Reading CSV files), all CSV files were read, with each file representing the author's publications throughout their academic career, as shown in Table 4.1. Table 4.1 presents fictitious raw data with no manipulation, although there are other variables not shown in this table. Additional variables available in the data include university affiliations per author, journal details for the published articles, descriptions and keywords of the publications, publication types, and sponsor information.

Thus, to read the files (over 400,000) in R (R Core Team, 2024), the *arrow* package was used, as it provides excellent speed and efficiency for reading and writing CSV files. At this initial stage, some adjustments were also made to facilitate the creation of the first concatenated dataset, such as selecting the variables to be used in the data analysis, filtering publications from 2000 to 2020, transforming the content of the CSV files into a single dataframe, and removing duplicate rows.

Table 4.1 – Example of fictitious authorship records from the SCOPUS database

EID	DOI	Title	...	Authors-id	City	Country	Year	Citation
x	111	...	...	1044	São Paulo	Brazil	2000	10
x	222	...	...	1044	São Paulo	Brazil	2005	5
x	333	...	...	1233; 1044	Paris; São Paulo	France; Brazil	2020	40

For the second step (Data reshaping), the observations needed to be structured by author and affiliation country, meaning each row in the dataset should represent a publication for an author with their affiliation country. This significantly increases the number of rows, as the number of authors in a single publication corresponds to the number of additional rows in the data. At this stage, we also filtered the publication types to include only "article" and "review". Additionally, it became possible at this step to classify the authors within a single publication. By the end of this step, authors could already be classified according to the types of academic mobility outlined by (SUBBOTIN; AREF, 2021).

The classification of researchers occurs in the third step, dividing them into six groups (Table 4.2). For researchers to be classified, at least two publications during the analyzed period were required. The authors' institutional affiliation country was considered their geographical location in the year the article was published.

Table 4.2 – Classification of researchers by mobility type

Classification	**Criteria**
Single Article	Author of a single publication
Non-Migrant	No evidence of international mobility
Immigrant	IA: other countries; FA: Brazil
Emigrant	IA: Brazil; FA: other countries
Return Migrant	IA: Brazil; TA: other countries; FA: Brazil
Transient Migrant	IA: other countries; TA: Brazil; FA: other countries

IA: Initial affiliation; FA: Final affiliation; TA: Transitional affiliation

In the fourth step (Adding academic journal metrics), the SJR (SCImago Journal Rank) metric was added to the final dataset as a *proxy* for the "quality" of the research published by Brazilian researchers. SJR data was extracted annually from the website[1] for all evaluated journals. These SJR values were then matched by year to each publication. Finally, the fifth (and last) step was used to remove unused variables,

[1] <https://www.scimagojr.com>

rename columns, and eliminate publications without an affiliation country. After completing all steps of data preprocessing and organization, variable transformations were performed to prepare for descriptive analyses.

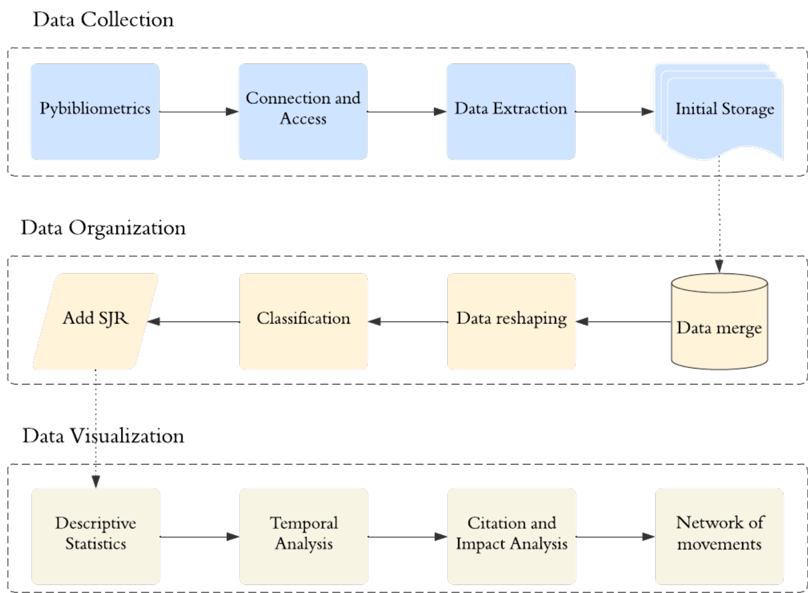
Before conducting the exploratory data analysis, the number of citations received was standardized based on the academic age of each publication (SUBBOTIN; AREF, 2021). This was achieved by calculating an annual citation rate, dividing the total number of citations a publication had received (as of November 2023) by its academic age. This procedure was used to eliminate the effect of time on the number of citations received by each publication. Additionally, a variable for "research collaboration" was created (sixth step). Scientific collaboration is often used in bibliometric studies to analyze the impacts of academic mobility (MARMOLEJO-LEYVA et al., 2015; LANCHO-BARRANTES; CANTÚ-ORTIZ, 2019). A commonly employed proxy for collaboration in these studies is co-authorship in scientific publications (AYKAC, 2021), which is also applied in this dataset. The collaboration variable is binary: it is assigned a value of 1 when there is at least one co-author with an international affiliation (non-Brazilian) and 0 when all authors have Brazilian affiliations.

At each stage highlighted here, we downloaded the generated data in .rda (R Data File) format for review and code corrections if necessary. The exploratory data analysis aims to examine and investigate the dataset, summarizing its main characteristics. During this analysis, descriptive statistics are calculated by author, affiliation (university and country), and mobility group. The variables used in this study mainly include the winsorized annual citation rate, the SJR index, and research collaboration. At this stage, differences among the analyzed variables by researcher group are identified through graphs and descriptive statistics. Finally, to address the questions posed in the "Introduction" regarding mobility patterns, maps are created to depict the origin and destination of mobile researchers. These maps are generated for emigrant and returning migrant researcher groups to study the main destinations of Brazilian researchers. Figure 4.1 provides a flowchart with the main steps for data collection, organization, and visualization.

4.3 Communication and Reproducibility

Quarto provides a unified framework for data science creation, combining code, results, and analyses to communicate with stakeholders (WICKHAM et al., 2023). Given the need for open access and technological advancements that enable the implementation of reproducible computational research workflows, it is crucial that practices ensuring reproducibility spread across various fields of knowledge (FIGUEIREDO et

Figure 4.1 – A framework of data acquisition and analysis



al., 2022). Reproducible research can also bring many benefits, such as enhancing the scientific value of the work, promoting equity in science, and advancing the dissemination of knowledge (DOGUCU, 2024).

Documents created in Quarto are fully reproducible and support a wide range of output formats, including PDFs, Word files, websites, books, presentations, and much more. The design of Quarto files allows for their use in three main contexts: (i) communication with decision-makers: aimed at those who want to focus on results and conclusions without needing access to the underlying code of the analysis; (ii) collaboration with other data scientists: providing both conclusions and the code that supports them, facilitating teamwork and serving as a future reference for the author; (iii) integrated research environment: functioning as a modern lab notebook, where it is possible to record not only what was done but also the reflections that guided the analytical process (WICKHAM et al., 2023).

This documentation is an excellent starting point for researchers looking to implement reproducible workflows in their projects. In the context of this thesis, all analyses and reports were developed in Quarto, with the source code available on Netlify, ensuring transparency and reproducibility.

Analysis of PhD Thesis

SECOND PART

Redigido conforme as normas das revistas X (Article 1) e *Scientometrics* (Article 2).

Academic mobility of Brazilian health and life sciences researchers: Analyzing international mobility of researchers by discipline using Scopus bibliometric data 2005–2020

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Abstract. Academic mobility plays a strategic role in strengthening research networks and advancing scientific progress, which relies on collaborative processes among researchers from different nations. In this study, bibliometric data from Scopus were used to investigate the academic mobility patterns of researchers in the health and life sciences fields, as well as their scientific productivity and the recognition of their publications. The results reveal that migrant researchers, although fewer in number, make significant contributions to the development of national science, with bibliometric indicators showing higher averages compared to non-migrant researchers. Additionally, the main destinations involved in the academic mobility of these health researchers were the United States, Canada, the United Kingdom, and France. Furthermore, researchers in the field of Medicine represented the largest group in the dataset and also the highest number of migrants. We observed that mobility enhances the performance and success of researchers by using propensity score matching to estimate causal effects and minimize confounding bias when examining the effect of mobility. By matching mobile researchers with non-mobile researchers with similar characteristics, we compared their scientific outcomes and found a positive impact of mobility on performance and scientific success in terms of citations received and the SJR indicator. The findings observed here aim to contribute to the advancement of studies on international academic mobility in Brazilian research, particularly regarding recognition, prestige, and the creation of collaboration networks.

Keywords: Scientific performance; Scientific success; Causal inference; Propensity score matching.

1 Introduction

In the globalized world, academic mobility and human migration must be extensively studied to understand national scientific systems. International academic mobility refers to the movement of researchers between institutions and countries, encompassing permanent migrations, temporary collaborations, academic visits, and participation in global projects. This phenomenon is essential for scientific progress as it facilitates knowledge exchange, the formation of collaborative networks, and innovation on a global scale (Robinson-Garcia et al., 2019). In this context, academic mobility and international collaboration are complementary dimensions of the internationalization process, with mobility serving as a catalyst for the establishment of global collaborations (Kato and Ando, 2017).

Academic mobility plays a strategic role in strengthening research networks and advancing scientific progress, which relies on collaborative processes among researchers from different nations. By enabling communication, cooperation, and the exchange of knowledge, mobility becomes essential for scientific advancement and the enhancement of researchers' performance (Momeni et al., 2022). However, mobility also presents challenges, such as inequalities in the distribution of human capital, which can weaken less advantaged regions while strengthening others. Recent studies have focused on exploring the different forms of academic mobility and their impacts on the development of national scientific systems (Doria Arrieta et al., 2017; Baruffaldi et al., 2020; Biglari et al., 2022). Academic mobility has shifted beyond the binary view of talent-winning ("brain gain") and talent-losing ("brain drain") countries, evolving into the concept of "brain circulation", which suggests that while researchers migrate from one country to another, all parties benefit (Aref et al., 2019).

Mobility analyses aim to uncover mobility patterns (Subbotin and Aref, 2021), differences in scientific productivity between mobile and non-mobile researchers (Aykac, 2021), the effects of mobility on research collaboration networks (Jonkers and Tijssen, 2008), and the impact of policies on the academic mobility phenomenon (Poitras and Larivière, 2023). Migration flows typically occur in a unidirectional manner, from developing countries to developed countries, as highlighted by Marginson (2006); Beine et al. (2008). Among the developing countries studied by some researchers, Brazil stands out as a nation that does not experience significant losses of scientific talent to other countries. This is primarily due to a predominant pattern of circulation, with researchers returning to Brazil after completing their proposed activities abroad (Rostan and Höhle, 2013; Lombas, 2017).

In this article, we focus on health and life sciences researchers in Brazil to study academic mobility. Research development in the health field is particularly important for addressing current public health challenges, such as emerging diseases, and for supporting governments in public policy formulation (Smith and Judd, 2020). In developing countries, the knowledge generated by scientific research enhances individual health conditions, thereby fostering the development of the population (Nundy et al., 2022). An analysis of Brazil's scientific and technological contributions revealed a significant role in combating the COVID-19 pandemic (Rosa et al., 2021).

Analyzing scientific mobility is a complex task due to the lack of adequate records and relevant migration indicators. Most studies conducted in Brazil are based on data from specific groups or programs (Lombas, 2017; McManus and Nobre, 2017). In recent years, academic mobility studies have increasingly relied on bibliometric data from databases like Scopus and Web of Science to infer the impact of country change on productivity (Aykac, 2021) and to analyze researchers' migration patterns over time (Subbotin and Aref, 2021). Thus, the aim of this study is to examine the dynamics and characteristics of mobility networks among Brazilian health researchers, as well as the impact of this mobility on national scientific production.

In this work, bibliometric data from Scopus was used to investigate the academic mobility patterns of health and life sciences researchers, as well as the scientific productivity and recognition of their publications. Additionally, this research aims to present mechanisms for data collection, detail the preprocessing and data cleaning steps, and visualize mobility patterns using thematic maps. Specifically, the research seeks to address the following questions: (i) What are the preferred migration flows of researchers? (ii) How do mobile researchers differ from non-mobile ones? and (iii) What is the impact of researcher mobility on national scientific production in the health field?

Thus, this study seeks to advance our understanding of the mobility patterns and characteristics of Brazilian health researchers. Finally, the remainder of the article is organized as follows: the "Related Work" section provides a literature review on academic mobility and its migration patterns. The "Data and Methods" section describes the steps for extracting, organizing, and visualizing the bibliometric data, as well as the process for estimating the causal effects of mobility. It also outlines the software and packages employed in each step. The research results are presented and discussed in the "Results and Discussion" section. The "Conclusion" section concludes the article with the main findings and the limitations of this study.

2 Related Work

Several studies examine academic mobility, its causes, and consequences using different databases, bibliometric variables, and countries of origin. The works of Jonkers and Cruz-Castro (2013) and Franzoni et al. (2014) addressed the topic of academic mobility with the aim of identifying the effects of researchers' international migration on their scientific productivity and collaboration networks. In both studies, these findings were based on questionnaires sent to specific groups of researchers, highlighting better performance among mobile researchers.

Focusing on specific groups of researchers, Lombas (2017) and Khakhuk et al. (2021) investigated the international migration of researchers using survey data. Lombas (2017) explored the international academic mobility of Brazilian researchers, discussing their international flows and trajectories by fields of knowledge. The data for this research were derived from questionnaires targeting researchers in Computer Science, Physics/Astronomy, and Economics, who received scholarships in the United States, France, and Great Britain between 1996 and 2007.

Meanwhile, bibliometric data from databases such as Scopus and Web of Science have seen expanded use in studies addressing aspects of researcher migration (Pranckutė, 2021). Research utilizing bibliometric data employs it in various ways to study characteristics of academic mobility, applying diverse statistical techniques and approaches to estimate the effects of researcher migration (Miranda-González et al., 2020; Aykac, 2021; Marini and Xu, 2023; Poitras and Larivière, 2023). For example, Miranda-González et al. (2020) used bibliometric data from Scopus to examine internal migration among researchers in Mexico, exploring the migratory movements of academics within the country. Similarly, Subbotin and Aref (2021) analyzed Scopus data to investigate international mobility patterns of Russian researchers and assess the impact of these movements on Russia's scientific system, both overall and within specific scientific fields.

In contrast, Aykac (2021) and Marini and Xu (2023) leveraged bibliometric data to estimate the effects on migrant and non-migrant researchers using regression models and synthetic control methods, respectively. However, while these studies utilize bibliometric data, they generally do not provide readers with details about the steps involved in data extraction (via web scraping), preprocessing, organization, and visualization. Therefore, this work aims to present mechanisms for analyzing researchers' international migration, starting from data acquisition through to statistical analysis and data visualization.

3 Data and Methods

The bibliometric data from the SCOPUS database is used to track the geographical location of Brazilian researchers in the health field, their publications, the "quality" of their published articles, as well as the citations they have received. The unique author identifiers (Scopus author IDs) and their affiliations will be used as the basis for identifying researchers based in Brazil. There is evidence that Scopus's author identification system is reliable for analyzing the migration and mobility of scientists (Aman, 2018), as almost all IDs correctly identify a researcher (Kawashima and Tomizawa, 2015), with 98.4% precision and 90.6% recall (Paturi and Loktev, 2020). Additionally, the Scopus database has extensive coverage of disciplines such as Natural Sciences, Medicine, Health Sciences, Engineering, and Technology (Pranckutė, 2021) and contains a larger number of Latin American journals in its core collection compared to other databases (Celeste et al., 2021).

The data covers the period from 2005 to 2020, assuming that the authors' institutional affiliation country corresponds to their geographic location in the year the article is published. The data was extracted from Scopus for authors with at least one Brazilian affiliation during the selected period, followed by the collection of all articles and reviews published by these researchers. As a quality indicator, the SJR (Scimago Journal Rank) index is used, representing a measure of the prestige of academic journals (McManus et al., 2020).

The number of citations received per article can represent scientific recognition (Reyes-Gonzalez et al., 2016), but it should be considered within the same scientific field for comparative analyses. In turn, the SJR is a bibliometric indicator capable of measuring the prestige or influence of a scientific journal through information obtained from the Scopus database (González-Pereira et al., 2010). Additionally, the SJR indicator is frequently used to represent the “quality” of scientific publications (Horta et al., 2020; Marini and Xu, 2023). Therefore, citation counts and the SJR constitute bibliometric measures commonly employed in research evaluation systems.

The sections below will outline the steps for data collection, processing, analysis, and the estimation of the causal effects of academic mobility on Brazilian researchers in the fields of health and life sciences.

3.1 Bibliometric data collection

The data collection process was carried out using the API (Application Programming Interface) provided by Scopus for data extraction, using the Python programming language. Specifically, we utilized the *Pybliometrics* library, which aims to assist researchers in bibliometric studies by offering a consistent and simple interface that automates data retrieval, JSON serialization, caching, and error handling (Rose and Kitchin, 2019). Furthermore, this library allows researchers to expedite their analyses in reproducible ways and enables scientific evaluators to automate the retrieval of assessment data.

The data collection was conducted in two stages. The first stage involved obtaining the seed data for Brazilian researchers, while the second focused on retrieving the publication histories of all authors in the initial dataset. In the first stage, the seed data collection consisted of identifying health and life sciences articles with contributions from authors affiliated with Brazilian institutions within the specified time period.

The API imposes collection limits, allowing each collection entity (controlled through a token) to extract only a restricted amount of data and perform a limited number of requests. Due to the large volume of data to be collected, the task was divided. Data collection was carried out by year and by field to obtain subsets of the data separately for subsequent processing. After the initial processing, the data (from the first collection) were organized into a database, initially using CSV files.

The processed data were then prepared for the second stage of collection. The preparation included: (i) extracting the IDs of Brazilian authors from the dataset; (ii) creating a dictionary linking institution IDs to countries; and (iii) removing duplicates and organizing the data for the continuation of the collection process.

The second stage involved gathering the publication histories of all authors present in the initial dataset. This process aimed to enable more extensive analyses in the future and to verify the completeness of the data collected in the previous stage. During this stage, articles were extracted from the Scopus database using each author's ID. The data were then organized by ID, generating a total of 435,391 files. Each generated file contains all publications associated with an author ID.

The publications extracted from Scopus correspond to the following areas within the domain of health sciences and life sciences: Agricultural and Biological Sciences; Biochemistry, Genetics and Molecular Biology; Dentistry; Health Professions; Immunology and Microbiology; Medicine; Neuroscience; Nursing; Pharmacology, Toxicology, and Pharmaceutics; Veterinary; and Multidisciplinary.

3.2 Data organization and analysis

The raw data is divided into several files, with each file containing all publications for a given author (author ID). The total number of files to be read and organized exceeds 400,000 CSV files, requiring the use of the *arrow* (Richardson et al., 2024), *readr* (Wickham et al., 2024), and *purrr* (Wickham and Henry, 2023) packages. After reading and concatenating the data into a single dataframe, it is necessary to classify authors into mobility groups, as described in Subbotin and Aref (2021). All the pre-processing, organization, visualization and estimation of mobility effects will be carried out in the R software (R Core Team, 2021).

The pre-processing and data organization stages were subdivided into the following steps:

1. Reading CSV files;
2. Data reshaping;
3. Classifying mobility groups;
4. Adding academic journal metrics;
5. Data cleaning and
6. Data analysis and visualization.

For the first stage, all the CSV files were read and concatenated into a single dataframe, with each observation representing the author's publications throughout their academic career, as shown in Table 1. Table 1 presents fictitious raw data with no manipulation, although there are other variables not shown in this table. Additional variables available in the data include university affiliations per author, journal details for the published articles, descriptions and keywords of the publications, publication types, and sponsor information. Thus, to read the files in *R*, the *arrow* package was used, as it provides excellent speed and efficiency for reading and writing CSV files. At this initial stage, some adjustments were also made to facilitate the creation of the first concatenated dataset, such as selecting the variables to be used in the data analysis, filtering publications from 2005 to 2020, transforming the content of the CSV files into a single dataframe, and removing duplicate rows.

For the second step (Data reshaping), the observations needed to be structured by author and affiliation country, meaning each row in the dataset should

Table 1: Example of fictitious authorship records from the SCOPUS database

EID	DOI	Title	...	Authors-id	City	Country	Year	Citation
x	111	...	...	1044	São Paulo	Brazil	2005	10
x	222	...	...	1044	São Paulo	Brazil	2005	5
x	333	...	...	1233; 1044	Paris; São Paulo	France; Brazil	2020	40

represent a publication for an author with their affiliation country. This significantly increases the number of rows, as the number of authors in a single publication corresponds to the number of additional rows in the data. At this stage, we also filtered the publication types to include only "article" and "review". Additionally, it became possible at this step to classify the authors within a single publication. By the end of this step, authors could already be classified according to the types of academic mobility outlined by Subbotin and Aref (2021).

The classification of researchers occurs in the third step, dividing them into six groups: (i) single article researcher (author of a single publication); (ii) non-migrant researcher (no evidence of international mobility); (iii) immigrant researcher (initial affiliation: other countries, final affiliation: Brazil); (iv) emigrant researcher (initial affiliation: Brazil, final affiliation: other countries); (v) returning migrant researcher (initial affiliation: Brazil, transitional affiliation: other countries, final affiliation: Brazil); (vi) transient migrant researcher (initial affiliation: other countries, transitional affiliation: Brazil, final affiliation: other countries). For researchers to be classified, at least two publications during the analyzed period were required. The authors' institutional affiliation country was considered their geographical location in the year the article was published.

In the fourth step, the SJR (SCImago Journal Rank) metric was added to the final dataset as a *proxy* for the "quality" of the research published by Brazilian researchers. SJR data was extracted annually from the website for all evaluated journals (SCImago, nd). These SJR values were then matched by year and ID journal to each publication.

In the fifth step was used to remove unused variables, rename columns, and eliminate publications without an affiliation country. After completing all steps of data preprocessing and organization, variable transformations were performed to prepare for descriptive analyses. At the end of each of the steps described above (steps 1 to 5), copies of the dataframe were saved in .rda (R Data File) format so that the complete data could be checked and adjustments made where necessary.

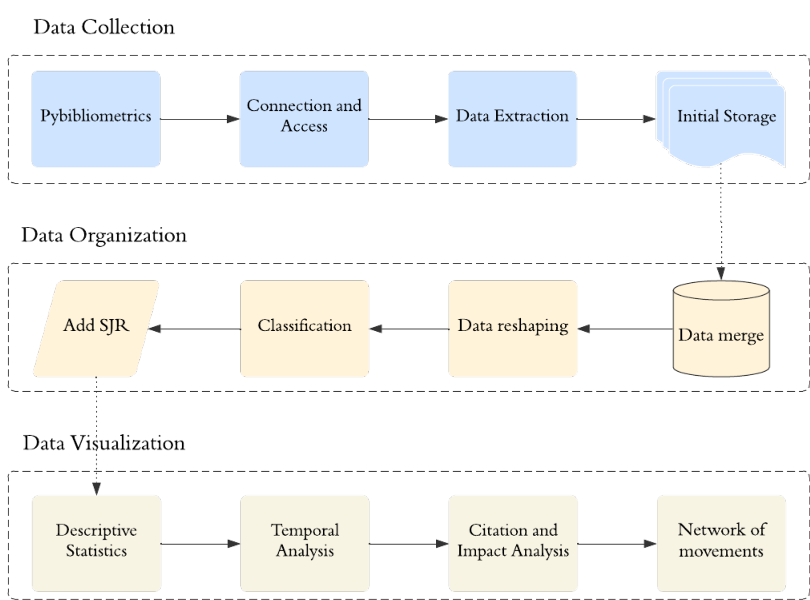
Additionally, the number of citations received was standardized based on the academic age of each publication (Subbotin and Aref, 2021). To do this, the annual citation rate was calculated by dividing the total number of citations a publication had received (as of November 2023) by its academic age. This procedure was used to eliminate the effect of time on the number of citations received by each publication. We also created a variable for "scientific collaboration", which is frequently used in bibliometric studies to analyze the impacts of academic mobility (Marmolejo-Leyva et al., 2015; Lancho-Barrantes and Cantú-

Ortiz, 2019). The collaboration variable is binary, where a value of 1 is assigned when there is at least one co-author with an international (non-Brazilian) affiliation and 0 when all authors have Brazilian affiliations.

The sixth step involved examining and investigating the dataset, summarizing its main characteristics through graphs and descriptive statistics. The variables used in this exploratory step primarily include the annual citation rate, the SJR index, and research collaboration. In this stage, differences between the variables analyzed by researcher groups are identified using graphs and descriptive statistics. To analyze academic mobility patterns, we created maps showing the main destinations of mobile researchers. These maps are generated for groups of emigrant and returning migrant researchers to study the main destinations of Brazilian researchers in the fields of health and life sciences.

A summary of the processes undertaken for data collection, organization, and analysis is presented in Figure 1. After all the procedures carried out to achieve a well-organized database, it was decided to exclude researchers with only a single article from the statistical analyses. The group of single-article researchers consists of 103,991 articles and 122,751 authors.

Fig. 1: A framework of data acquisition and analysis



3.3 Analysis of causal effects of mobility

To estimate causal effects in observational studies, where treatment assignment is not random, it is necessary to use techniques that account for differences between the treated and control groups. The Propensity Score Matching (PSM) method is employed to correct sample selection bias due to observable differences between treatment and control groups (Dehejia and Wahba, 2002). This method is widely used to evaluate the economic impacts of non-random treatments (Bertram-Huemmer and Kraehnert, 2018; Seriño et al., 2021), as well as in studies examining the effects of academic mobility on research performance (Momeni et al., 2022; Jing et al., 2023).

The PSM technique was developed by Rosenbaum and Rubin (1983) to measure the conditional probability of assignment to a specific treatment given a vector of observed covariates. In other words, it is used to estimate the effects of a treatment on a given group. Formally, Rosenbaum and Rubin (1983) define the conditional probability of assignment to treatment ($A = 1$), given the observed covariates X, for N units, as the following expression:

$$e(x) = P(A = 1|x) \tag{1}$$

in which it is assumed that

$$P(A_1, ..., A_n|x_1, ..., x_n) = \prod_{i=1}^{N} e(x_i)^{A_i}\{1 - e(x_i)\}^{1-A_i}.$$

The function $e(x)$ is called the propensity score, that is, the probability of receiving the treatment ($A = 1$), given the observable variables x. Therefore, this function can be interpreted as the likelihood of each unit receiving the treatment, given the variables already observed. In this regard, Rosenbaum and Rubin (1983) point out that in non-randomized experiments, the function $e(x)$ (Equation 1) is almost always unknown and should be estimated from the observed data using some model, such as the *logit* model.

Rosenbaum and Rubin (1983) highlighted some extremely important assumptions for generating good results for the PSM model, which should not be overlooked, namely (i) balancing score and (ii) strong ignorability. The assumptions of propensity score matching are necessary for making acceptable comparisons between the treated and control groups, as these groups come from non-randomized experiments.

The first assumption is used to make comparisons between the treated group and the control group more appropriate. It indicates that the probabilities of each unit receiving or not receiving the treatment should be the same. Rosenbaum and Rubin (1983) defined balancing scores, $b(x)$, as a function of observed covariates X, where the conditional distribution of x given $b(x)$ is the same for the treated group $A = 1$ and the control group $A = 0$.

The second assumption for the PSM model, strongly ignorable, is based on two concepts: (i) the conditional independence assumption (CIA) and (ii) common support. The conditional independence assumption indicates that the potential outcomes must be independent of the treatment, conditional on the observed variables. The common support assumption aims to make two distinct groups comparable, as it ensures that the treatment groups have matches in the control group, based on the selected covariates x (Rosenbaum and Rubin, 1983).

In this study, the treatment refers to academic mobility, meaning that researchers exposed to mobility receive $A = 1$, while those not exposed receive $A = 0$. The covariates (X) used to control for the effects refer to the characteristics of the researchers (number of publications, career age, and collaborations) and their scientific performance before mobility (number of citations and SJR). The implementation of PSM was performed for the emigrant and return migrant groups using 1:1 nearest neighbor matching.

To assess whether the units (researchers) being compared satisfy the assumptions of this model, the overlap of probability distributions and the standardized mean difference (SMD) were evaluated. Additionally, the differences in the means of the treated and control groups for the analyzed variables (citations and SJR) were tested using the t test for two independent samples.

4 Results

In this section, we present the main results of our analysis. First, we explore citation-based performance among researchers according to their mobility type. Next, we discuss the geography of mobile researchers and analyze their migration flows. Then, we compare the performance of immigrants and emigrants based on the citations they received (normalized by publication years). In the final part of the results section, we assess the overall impact of migration on the Brazilian scientific system.

4.1 General Results

The average number of publications for all authors, without distinguishing mobility groups or fields of knowledge, is 8.39 publications during the period from 2005 to 2020. Table 2 presents the general results regarding the number of published articles, the total number of authors, and the relative frequency of authors (%) for each type of academic mobility during the period from 2005 to 2020. The non-migrant group stands out as the most representative, with 566,828 articles published by 182,075 authors, accounting for 85.01% of all researchers. In contrast, the migrant groups show lower representativeness: immigrants account for 2.18% of authors, emigrants for 3.46%, return migrants for 7.35%, and transient migrants for 2.04%. These data emphasize the predominant role of non-migrants, although migrant authors also make substantial contributions to academic production.

In Figure 2, we explore the variation in the number of publications by type of mobility. The x-axis indicates the publication ranges, and the last value (100+) represents the range of researchers with more than 100 publications throughout the analyzed period. The non-migrant group dominates most of the publication

Table 2: General results by mobility type

	N. articles	Authors	Frequency (%)
Non-migrant	566,828	182,075	85.01
Immigrant	91,754	4,663	2.18
Emigrant	117,214	7,407	3.46
Return migrant	356,101	15,742	7.35
Transient migrant	258,205	4,376	2.04

ranges, indicating that most researchers in each range do not have a migration history. However, the proportion of non-migrants tends to decrease in the higher publication ranges. This suggests that migrant researchers may be more productive or more involved in larger-scale projects (Jonkers and Tijssen, 2008).

As the number of publications increases (horizontal axis in Figure 2), the contribution of the immigrant, return migrant, and transient groups grows. This may indicate that researchers exposed to academic mobility (migrants) are more represented among those with more prolific careers, and possibly, migration is associated with international collaborations or access to resources that increase productivity (Jonkers and Cruz-Castro, 2013). The "transient migrant" group has the highest proportion in the extremes of publication ranges, suggesting that researchers with temporary experiences in other countries may be part of highly productive scientific networks. In turn, return migrants also appear more frequently in the higher ranges. This may reflect a "return effect", where researchers who migrated and then returned bring back accumulated experience and resources (Aykac, 2021).

The academic mobility of Brazilian researchers in the fields of health and life sciences is detailed based on their main area of expertise (field of knowledge of the publication) in Figure 3. Researchers are divided into non-migrants and migrants (groups exposed to academic mobility), as shown in Figure 3. Regarding fields, medical articles were the most prevalent in the dataset from 2005 to 2020, also indicating a higher number of migrants. Next, the fields with the most publications are Agricultural and Biological Sciences (AGRI) and Biochemistry, Genetics, and Molecular Biology (BIOC). It is worth noting that publications can have multiple classifications of areas, which increases the total number of publications by repeating the same work in different fields. When specifically analyzing researchers in Medicine, due to the higher concentration of migrants in this field, we observe that most of them move to the United States, Canada, and the United Kingdom.

The results shown in Table 3 reveal the number of authors by field and type of academic mobility. These results complement the information in Figure 3, pointing to the higher number of researchers in the field of Medicine across all types of academic mobility, but also indicating that most of the migrants are temporary (return migrants). Similarly, in the second field with the highest number of authors (BIOC), return migration is also observed, as well as in the other fields analyzed. The results found by field can reveal important information about the

Fig. 2: Percentage of researchers by type of mobility and range of publications

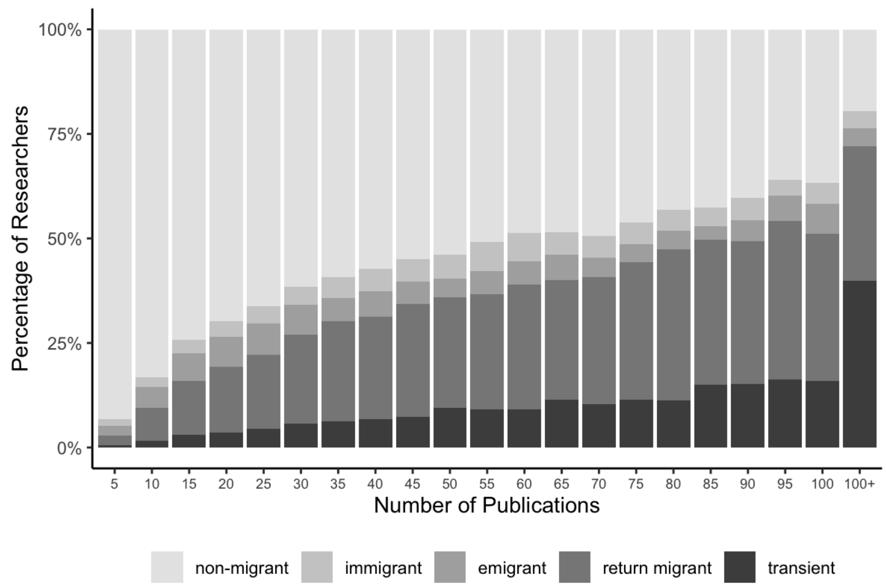
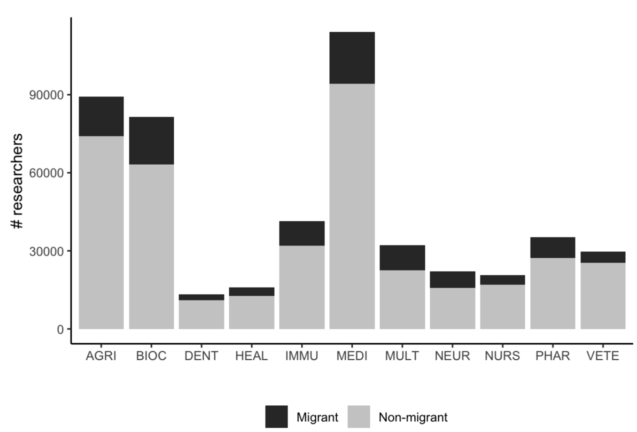


Fig. 3: Number of Brazilian health researchers by area



return of qualified researchers for national development and advancements in health research.

In order to gain a more in-depth understanding of the movements of Brazilian health and life sciences researchers, we carried out a detailed analysis of the most frequent origins and destinations throughout the database. This mapping allowed us to identify the main flows of scientific mobility, highlighting the countries that

Table 3: Number of authors by type of mobility and area

Area	Non-migrant	Immigrant	Emigrant	Return migrant	Transient
DENT	10,995	246	419	1,430	200
HEAL	12,682	428	505	2,115	256
IMMU	32,019	1,070	1,748	5,578	962
MEDI	94,210	2,593	3,982	10,765	2,488
MULT	22,596	1,156	1,716	5,649	1,018
NEUR	15,777	715	1,185	3,799	602
NURS	16,908	405	533	2,454	326
PHAR	27,341	831	1,360	5,007	693
VETE	25,381	520	806	2,691	341
AGRI	74,173	2,069	3,132	8,309	1,679
BIOC	63,254	2,188	3,746	10,289	1,984

most attract Brazilian researchers and those that receive them back. In addition, we segmented this data to focus on two specific groups: emigrants, who leave Brazil in search of opportunities abroad, and return migrants, who return to the country after a period of international work. In this way, we sought to identify trends, challenges and impacts related to academic mobility in the context of health.

4.2 Flows, origins, and destinations

Figure 4 represents the distribution of mobile researchers based on their mobility score. The mobility score ranges from 1 to 24, with a decreasing number of researchers as mobility scores increase. Most mobile researchers ($\approx$83%) have only one movement throughout their careers.

We defined the academic origin country (destination) as the country that appears in the first (last) publication of an individual researcher. Figure 5 illustrates the most common countries of primary affiliation associated with the first publications of Brazilian health researchers (in Fig. 5a) and their last publications (in Fig. 5b) from 2005 to 2020. In Figure 5, we show the origin and destination countries other than Brazil to observe the countries associated with the academic mobility of Brazilian health researchers. This was necessary because most researchers are non-migrants. The countries with the highest destinations for health researchers are primarily from North America and Europe. These results were also found in McManus et al. (2020), which highlighted the strong collaboration of Brazilian authors with the United States, the United Kingdom, Spain, France, Germany, Italy, Canada, Portugal, and Australia.

We also investigated the international trajectories for two specific groups of researchers: emigrant researchers and return migrant researchers, which are presented in Figures 6 and 7. Figure 6 illustrates the international trajectories

Fig. 4: Distribution of mobile researchers based on their mobility score

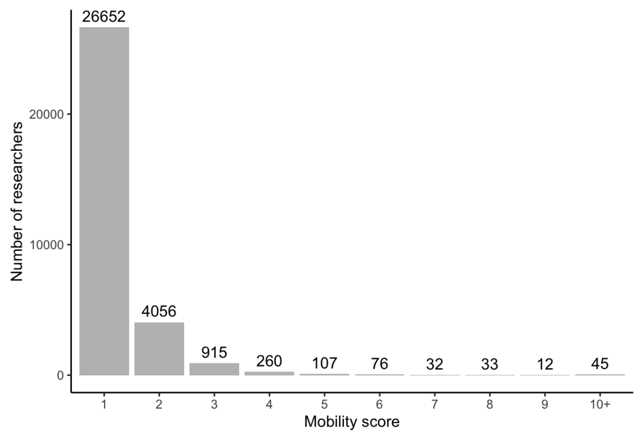
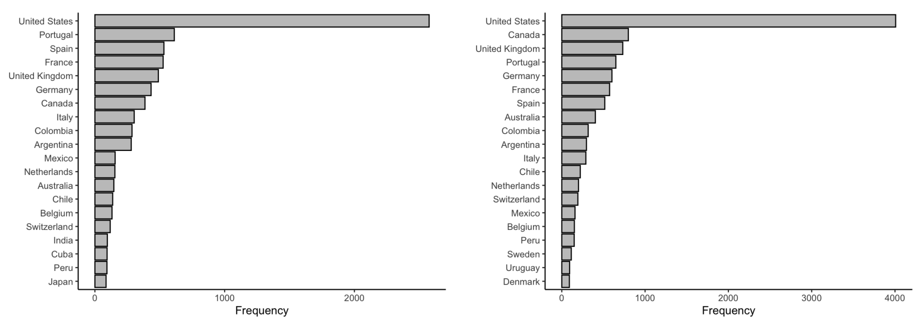


Fig. 5: Twenty most common countries of academic origin (a) and academic destination (b)



(a) (b)

of researchers who moved out of Brazil from 2005 to 2020, revealing the United States as the primary destination (represented with the thickest line). In addition to the movements to the United States (2574 movements), we observed migration flows to Europe (United Kingdom, Germany, Portugal, and France) and East Asia. Some Latin American countries also received a significant flow of Brazilian health researchers, such as Colombia, Chile, Peru, and Argentina.

From the perspective of return migrants (Figure 7), the United States (5314 movements), France (1052 movements), and the United Kingdom (1034 movements) were the most popular countries chosen by researchers for temporary mobility before returning to Brazil. The main countries involved in the migration flows in Figure 7 (USA, UK, and France) were also observed in studies of the Brazilian government's "Ciência sem Fronteiras" mobility program, which provided scholarships to university students in various fields of knowledge and study levels (undergraduate, PhD, postdoctoral, etc.) (McManus and Nobre, 2017; Sandes-Guimaraes et al., 2020). Furthermore, the total migration flows to these three most frequent destinations were more than 50% higher than those to other countries.

Fig. 6: Network of movements of Brazilian health researchers (Emigrants)

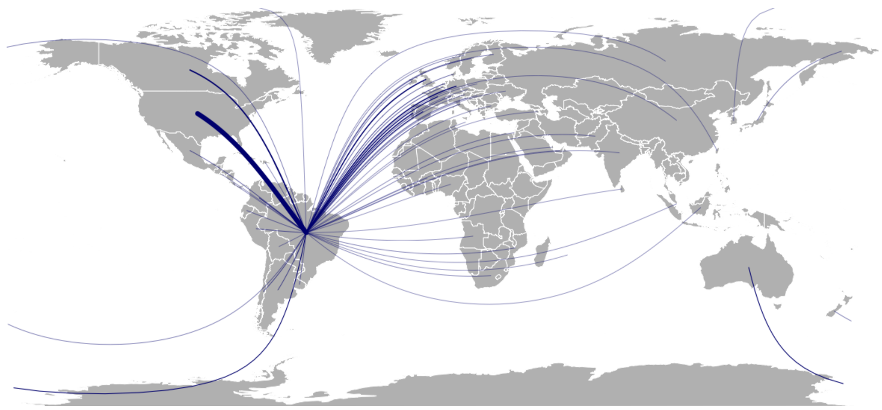
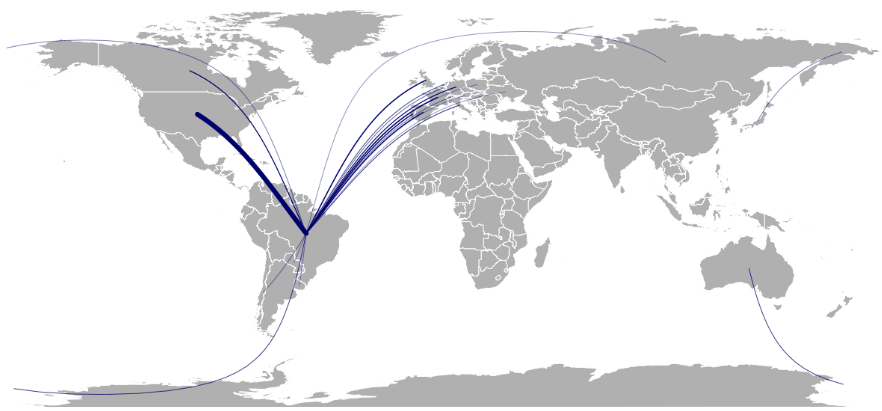


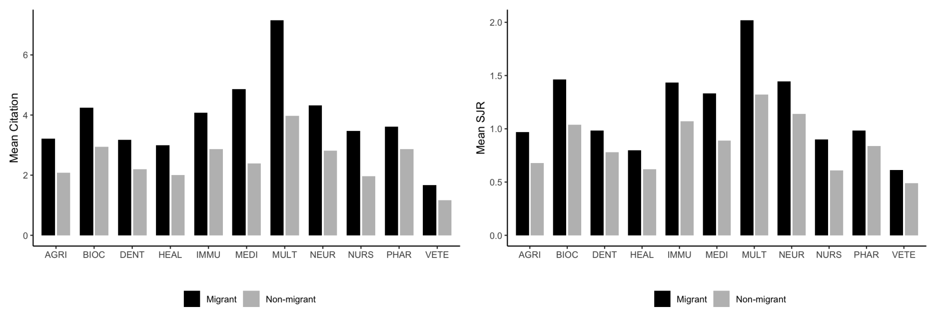
Fig. 7: Network of movements of Brazilian health researchers (Return migrants)



4.3 Assessment of the impact of mobility

To investigate the effects of mobility on the recognition and prestige of the analyzed publications, we created Figure 8 to present citation and SJR measures. The results indicate that average citation rates are generally higher for migrants (authors exposed to academic mobility) compared to non-migrants (Figure 8a). Notably, the Multidisciplinary field (MULT) stands out with the highest average citation rate. Similarly, publications in this field also exhibit the highest average SJR rate (Figure 8b). Among the main fields of study, substantial differences were observed in the average citation rates received by researchers, with multidisciplinary researchers achieving higher rates (Subbotin and Aref, 2021).

Fig. 8: Averages of predictor variables by mobility group



(a) (b)

In Table 4, we present the means of the analyzed variables for the different groups of researchers, using superscript letters to indicate significant differences. Treatments (researcher groups) with different letters are significantly different at the 5% level. The results reveal substantial disparities in citations between mobile and non-mobile Brazilian health researchers. The average citations for non-migrants were significantly lower than those for the other researcher groups. These citation disparities between mobile and non-mobile researchers have also been observed in studies on the academic mobility of Russian researchers (Subbotin and Aref, 2021) and Turkish researchers (Aykac, 2021).

Disparities in scientific performance in terms of "quality" measured by the SJR variable, also suggest that Brazilian health researchers exposed to international academic mobility exhibit higher averages than those not exposed to the phenomenon. The SJR averages are notably higher for transient and emigrant researcher groups. Furthermore, Table 4 shows that international collaborations differ significantly between non-migrants and the other groups. However, the immigrant, emigrant, and returnee groups show similar rates of international collaboration. Overall, although researchers with international mobility represent only a small fraction of all researchers in the sample (9.55%), they made essential contributions to the Brazilian scientific system, as evidenced by the substantial number of citations they received and the journal rankings reflected by the SJR.

To estimate the effect of mobility considering the covariates, we used propensity score matching (PSM), which calculates the causal effects of the treatment

Table 4: Average and standard deviation of bibliometric data by mobility type

	Citation	SJR	Collaboration
Transient migrant	6.75 ± 24.04	1.96 ± 2.35	82.05 ± 141.00
Emigrant	5.13 ± 16.59	1.63 ± 2.12	15.52 ± 29.18
Immigrant	3.97 ± 11.68	1.27 ± 1.52	13.50 ± 29.70
Return migrant	3.45 ± 12.48	1.13 ± 1.32	14.56 ± 22.74
Non-migrant	2.35 ± 5.88	0.79 ± 0.78	1.59 ± 4.12

(academic mobility). We selected five covariates for matching (Table 5). The variables Citation$_{t-1}$ and SJR$_{t-1}$ represent the averages of these respective variables for each researcher before mobility, to obtain a control group more similar to the treated group. We then compared the scientific outcomes regarding citations and SJR for these two groups (emigrant and return migrant). To control for the effect of mobility by group, we analyzed emigrant and return migrant mobile researchers and paired them separately with non-mobile researchers.

Table 5 presents these covariates and the Standardized Mean Difference (SMD) to evaluate the balance of the covariates before and after matching. The covariates used here also served to control for the effects of mobility in Momeni et al. (2022). Stuart et al. (2013) recommended a threshold of 0.1 to confirm the balance of covariates between two groups. All SMDs are below the threshold after matching, which demonstrates that all covariates were balanced. Table 6 presents the results of the t tests, which reveal higher citation and SJR values for mobile researchers, both emigrants and return migrants.

Table 5: The Balance report before and after matching treatment (emigrant and return migrant) and control (non-mobile) groups

	Emigrant		**Return migrant**	
	Before Matching	After Matching	Before Matching	After Matching
	SMD	SMD	SMD	SMD
Covariate:				
# of articles	0.31	-0.03	0.59	0.04
Career age	0.54	-0.08	1.20	0.02
Collaboration	2.12	0.04	1.09	-0.06
Citation$_{t-1}$	0.15	-0.04	0.13	-0.03
SJR$_{t-1}$	0.42	-0.03	0.44	0.02
Sample sizes:				
Control	182,075	6,479	182,075	14,308
Treated	7,407	6,479	15,742	14,308
Unmatched	-	928	-	1,434

Table 6: Comparison of Means for Emigrant and Return Migrant Groups

	Emigrant			**Return Migrant**		
	Mean Control	Mean Treated	t-value	Mean Control	Mean Treated	t-value
Citation	3.00	4.07	-13.73***	2.56	3.19	-13.13***
SJR	1.10	1.34	-14.60***	1.02	1.18	-19.43***

Significant codes: $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5 Discussion

6 Conclusions

This study was designed to investigate the patterns of academic mobility among health researchers, as well as the scientific productivity and recognition of their publications, based on bibliometric data from Scopus. Existing works in the literature have already highlighted the positive effects of academic mobility on scientific production, as well as its impact on research collaboration networks. However, few studies have relied on the use of bibliometric data to investigate the impacts of academic mobility and explore the mobility patterns of Brazilian health researchers.

The results observed here aim to contribute to the advancement of studies on international academic mobility in Brazilian research, particularly concerning recognition, prestige, and the creation of collaboration networks. In this study, we aimed to detail the methodological processes necessary for data collection, analysis, and visualization, to make it possible to apply these processes in other bibliometric studies in different fields and countries. The analyses were conducted for all health researchers in general, followed by results based on flows, origin and destination, and finally, researchers divided by fields, considering the time window from 2005 to 2020.

The findings reveal that migrant researchers, although fewer in number, significantly contribute to the development of national science, with bibliometric indicators showing higher averages compared to non-migrant researchers. Furthermore, the main destinations involved in the academic mobility of these health researchers were the United States, Canada, the United Kingdom, and France. In addition, researchers in the field of Medicine represented the largest group in the dataset and also the highest number of migrants.

This study acknowledges certain limitations. Bibliometric data were employed to assess authors' mobility and career age, with Scopus author IDs used to link researchers to their publications. While the Scopus author identification system is highly precise in associating articles with specific IDs, it may not comprehensively account for an author's full body of work, occasionally generating multiple IDs for the same individual (Moed et al., 2013). Furthermore, the analysis could benefit from additional covariates, such as gender, the GDP of the destination country, and specific scientific subfields, which were not included in this study.

Finally, it is important to highlight that the findings in this study provide valuable information for the Brazilian government, which can be used in the formulation of public policies for international academic mobility. The results point to advantages in scientific production generated by researchers exposed to academic mobility, which could benefit the development of national science, particularly in the health sector. It is also worth mentioning that the Brazilian government is currently focused on repatriating highly qualified Brazilian individuals through the "Talent Repatriation Program - Knowledge Brazil". This federal policy aims to attract and retain qualified human resources in the country starting in 2024. It would also be beneficial for the development of national science to encourage local researchers to gain greater scientific impact experience abroad (including scholarships, grants, or awards) that specifically benefit the most productive scientists upon their return.

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Authors' Contributions

Leonardo Biazoli: Conceptualization, Methodology, Formal analysis and Writing - Original Draft.
Priscila Costa Albuquerque: Conceptualization, Methodology and Writing - Review & Editing.
Bruna de Paula Fonseca: Conceptualization, Methodology and Writing - Review & Editing.
Eric Fernandes de Mello Araújo: Conceptualization, Data Curation, Methodology and Writing - Review & Editing.
Izabela Regina Cardoso de Oliveira: Conceptualization, Methodology and Writing - Review & Editing.

Conflict of interest statement

The authors have no competing interests.

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Data availability

The codes used in this work, including raw data download, data processing, descriptive analysis and data visualization are publicly available on *Github* at https://github.com/leobiazoli/thesis.

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International Mobility Boosts Scientific Careers: A Synthetic Control Analysis of Brazilian Researchers

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Abstract. The study of the relationship between scientific output and academic mobility is crucial for the formulation of public policies in developing countries. In this article, we present an application of the generalized synthetic control method for mobile Brazilian researchers in the health and life sciences field to estimate the effects of mobility on the recognition and prestige of publications, as well as on international collaboration networks. To improve the necessary assumptions prior to implementing the synthetic control and to enhance estimation accuracy, we introduce a matching technique as a pre-selection step for the synthetic control design, pairing control individuals equivalent to treated individuals based on pre-intervention observable variables. Finally, we apply the generalized synthetic control to estimate the causal effects of mobility on the scientific output of mobile Brazilian researchers, comparing the impact on the number of citations received, SJR, and international collaborations after the first year of mobility with their respective synthetic authors. The results suggest that international mobility leads to changes in the analyzed variables, particularly for emigrant mobile researchers. This work provides guidance on the use of synthetic control in bibliometric data. Additionally, the study offers insights for the development of science internationalization policies. The method we employed can also be applied to study the effects of international mobility in other fields of knowledge.

Keywords: Bibliometrics data; Synthetic Control; PSM; Brazilian science; Academic Mobility.

1 Introduction

The international mobility of researchers is an event that must be considered in the globalized world, since it is capable of promoting both brain drain and brain gain in terms of countries' human capital. This means that countries can attract highly qualified professionals through this event, but they can also lose them. In particular, several studies suggest that this event can affect the socio-economic and innovative development of these countries, as well as posing challenges for researchers and policymakers (Beine et al., 2008; Jonkers and Tijssen, 2008; Subbotin and Aref, 2021).

Studies on the phenomenon of brain drain were developed for different areas of knowledge (Hussey, 2007), countries (Khakhuk et al., 2021) and methodological approaches (Xue et al., 2021). To study the causes and consequences of brain drain in Brazil, for example, Khakhuk et al. (2021) found a relationship between the phenomenon and the motivation for higher salaries and opportunities for academic development. Lombas (2017) studied the paths taken by Brazilians towards academic internationalization, finding that the country does not suffer substantial losses of scientific talent abroad, with the predominant movement being circulation.

The methodological approaches used to quantitatively estimate the effects of international mobility on some variable of interest represent a variety of models, in particular, quasi-experimental models. Such models serve to evaluate the effects of an intervention on the result of a variable of interest, especially when the assignment of the experiment does not occur randomly (Cook et al., 2002). Recent studies have used quasi-experimental methods to estimate the causal effects of mobility on scientific productivity and research collaboration, applying the regression discontinuity approach (Baruffaldi et al., 2020; Xue et al., 2021), differences in differences (Doria Arrieta et al., 2017) and synthetic control method (Aykac, 2021).

The synthetic control method seeks to determine the causal relationship in observational studies by constructing synthetic units that are compared with the actual treated units (Abadie, 2021). This approach can be used in various contexts, including studies in the science of science (Farys and Wolbring, 2017; Aykac, 2021; Zullo and Churkina, 2023). Furthermore, a generalization of this method was recently developed to accommodate multiple treated units and different intervention periods (Xu, 2017), which can also be applied in studies of researchers with different types and periods of academic mobility.

As we can observe, evaluating the effects caused by the international mobility of researchers remains a challenging task that can be evaluated from different perspectives. This study aims to analyze the causal effects of international mobility on the careers of Brazilian health and life sciences researchers, using data from publications collected in Scopus. We propose a methodological approach in this study using the generalized synthetic control method to investigate the possible effects of international mobility, which can be applied to different intervention periods, as in the case of various years of departure of Brazilian researchers. As part of this proposal, we also applied a matching method, such as propensity score matching, to select the donor sample used in the model. The application of this methodological approach is unprecedented for bibliometric data on Brazilian researchers, aiming to contribute to causality studies for this group of researchers and advance research in other fields of knowledge.

The publications of Brazilian researchers are extracted from the SCOPUS database and then used to track the authors' information, such as country of affiliation, number of citations received per article, and their collaborations. In addition, SJR (Scimago Journal Rank) information is used as a proxy for the quality of scientific articles. Based on the authors' affiliations, we will classify them according to the type of academic mobility. The final dataset covers the time window from 2000 to 2020 for Brazilian scientific articles, classified into groups of international mobility (non-migrants, immigrants, emigrants, return migrants, and transients). Careful pre-processing was conducted on the data to reduce potential biases inherent in this type of study.

The remainder of the article is structured as follows: the "Academic Mobility Analysis" section provides an overview of the literature on the relationship between scientific mobility and scientific production, as well as causal inference. The "Design and Data" section addresses each of the chosen variables and also details the implementation of the synthetic control model. It also describes the data and methodology used. The research results are presented and discussed in the "Results and Discussion" section. The "Conclusion" section concludes the article with the main findings and limitations of this study.

2 Academic Mobility Analysis

Assessments of the effects of brain drain can be carried out from different perspectives and using different statistical models. One of the challenges in dealing with the assessment of international brain migration refers to the endogeneity problem, which arises when unobserved variables that also affect the response variable are not identified (Ngoma and Ismail, 2013). Furthermore, to study this topic, some works address the use of multiple techniques to evaluate the effects of brain drain on some variable of interest (Ngoma and Ismail, 2013; Xue et al., 2021).

Subbotin and Aref (2021) studied international mobility in academia, focusing on the migration of researchers from Russia. The data they use refers to a database with more than 2.4 million publications from Scopus, analyzing all researchers who published with a Russian affiliation address in sources indexed in Scopus from 1996 to 2020. Given this scenario, the approach adopted by the authors serves as a good reference for the problem of *big data* structures to analyze academic migration in other countries.

Baruffaldi et al. (2020) examined the impact of a grant program that promotes international mobility on scientific results and researchers' careers. Causal estimates were obtained using discontinuous regression and it was found that the fellowship effectively supports research periods abroad that often extend beyond the duration of the fellowship, without increasing the likelihood of permanent migration. Furthermore, the research revealed that researchers awarded the grant increase the quality of their production, although the effect on the quantity of production and careers is not significant.

Ngoma and Ismail (2013) empirically analyzed the impact of brain drain on investment in human capital in migrants' countries of origin, using regression models and instrumental variables. In the results obtained by the authors, it was evident that the probability of skilled migration has a negative impact, in the short term, on investment in human capital at secondary and higher education levels, while the long-term incentive effect is statistically insignificant.

The effects of the brain drain were also evaluated by Doria Arrieta et al. (2017), whose objective was to estimate the impacts of this brain drain on European science. To do this, they applied two causal inference models, synthetic control and difference-in-differences, to data from millions of academic publications from 1996 to 2012, which are disaggregated into 14 thematic areas and 32 European countries. The authors observed a negative impact on cross-border scientific collaboration in European countries, highlighting the need for effective incentives and policies for return to the country of origin.

The causes and consequences of academic migration of researchers from Brazil, Russia, India, China, South Africa (BRICS) to developed countries were studied by Khakhuk et al. (2021), seeking to find reasons for this migration and identify the consequences of process through the example of a Russian university, studying the characteristics of personal experience and motivation of students and teachers. The methodological approach addressed in this research was quantitative using questionnaires, resulting in the most significant motivation for academic migration being the expectation of higher income, social protection and access to academic development opportunities.

Aykac (2021) studied the effects of the mobility of Turkish researchers on the quality of academic publications, measuring changes in citation, impact factor, collaboration networks, and citations by other foreign researchers. To estimate the effects of international migration, the author used the synthetic control method, comparing mobile Turkish researchers with their non-mobile counterparts. Differences were observed in scientific productivity, the "quality" of journals in which articles were published, and collaboration networks between researchers who left Turkey and those who did not migrate.

In the study by Aykac (2021), the causal impacts of academic mobility were examined using the synthetic control method to compare two groups of Turkish researchers: those who moved to the United States and remained there and those who eventually returned to Turkey. To perform these analyses, the year of each researcher's first U.S. publication was standardized as the treatment year, and a synthetic control was created for each individual researcher, utilizing a donor pool of over 79,000 researchers. The complexities involved in creating synthetic controls could be mitigated by applying generalized synthetic control, allowing for multiple treated units and treatment periods.

In summary, the works discussed here involving international brain migration have used quasi-experimental models, in particular regression discontinuity, differences-in-differences and synthetic control. The following sections discuss the causal effects and specificities of the synthetic control methods that will be considered in this paper.

3 Causal Inference

Randomized experiments represent the gold standard in statistical analyses (Rubin, 2008). However, in social experiments, it is not always possible to use randomization to study certain phenomena, making it necessary to use observational data. Causal inference encompasses a large number of procedures to deal with non-experimental data, such as regression-based models, regression discontinuity designs, difference-in-differences, and synthetic control methods. The difference-in-differences methodology is a version of fixed-effects estimation using aggregated data, meaning the causal effect of the treatment is estimated as a fixed parameter (Angrist and Pischke, 2009). However, difference-in-differences methods use a single control unit or a simple average of control units, requiring the parallel trends assumption.

Although the choice for randomized experiments should be made whenever possible, in several situations it is not viable, such as social science events (Rubin, 1974). Thus, an alternative to the problem is to use estimates of the causal effects of the treatment in the non-random event, through a quasi-experimental model. The estimation of causal effects occurs with the identification of a certain event, which exerts a change in the variable of interest, when compared to a control group. The central problem in identifying a causal effect is that the variable of interest is observed under either treatment or control, but not under both factors (Dehejia and Wahba, 2002).

Considering that W is the treatment and Y the result, formally, Y^W refers to the potential result in the presence and absence of treatment. The binary variable W is responsible for indicating the units that participated in the treatment ($W = 1$) or did not participate in the treatment ($W = 0$). Therefore, Y^1 is the value of the variable of interest when the unit is subjected to the treatment, and Y^0 is the value of the variable of interest when the unit is not subjected to the treatment. According to Heinrich et al. (2010), the observed result can be denoted as:

$$Y = (1 - W)Y^0 + WY^1. \tag{1}$$

Therefore, based on the potential results, the causal effects of the treatments can be estimated. According to Rosenbaum and Rubin (1983), the average treatment effect (ATE) is given by

$$ATE = E(Y^1) - E(Y^0), \tag{2}$$

where $E(.)$ indicates the expectation of the population of interest. The Equation 2 is the difference between the effect of the treatment for the treated population and the effect of the treatment on the untreated population, therefore, it is not possible to know $E(Y^0)$ and $E(Y^1)$ in the same unit.

Heinrich et al. (2010) also highlight the average effect of the treatment on the treated (ATT), which measures the effect of the treatment on the units that participated in the treatment, that is,

$$ATT = E(Y^1 - Y^0 | W = 1). \tag{3}$$

In this case, through the treated population, there is the scenario in which this population received the treatment ($W = 1$) and did not receive it ($W = 0$), but, again, it is not possible to obtain data for the situation of what would have happened without the treatment in the same population, that is, the counterfactual. Faced with such a situation, a question to ask is how to use observational data to link observed results to potential outcomes? Data from observational (quasi-experimental) studies can be used to provide estimates of what the response variable would be like in the absence of treatment. This can be done differently between quasi-experimental models, as well as whether a control group is used or not.

3.1 Synthetic control methods modeling

The synthetic control method (SCM) is a widely used methodology to determine the causal relationship in observational studies (Abadie, 2021). This approach was developed by Abadie and Gardeazabal (2003) with the aim of constructing the synthetic trajectory of the variable of interest for treated units over time, based on multiple control units. In this initial approach, Abadie and Gardeazabal (2003) used SCM to assess the economic effects following the terrorist attacks in the Basque Country. The synthetic control method has applications in various observational studies, involving evaluations of public policies on tobacco control (Abadie et al., 2010), effects of political regimes on infant mortality (Pieters et al., 2016), economic impacts caused by conflicts in Turkey (Bilgel and Karahasan, 2019), economic impacts of natural disasters on election outcomes (Masiero and Santarossa, 2021), and analyses of vaccination incentive policies (Sehgal, 2021).

Formally, let $J + 1$ be the number of units, where the first unit is exposed to the treatment of interest, thus having J units as potential controls (Abadie, 2021). Let Y_{it}^N be the response variable that would be observed in the absence of the treatment, for units $i = 1, ..., J + 1$ and in the time period $t = 1, ..., T$. Let T_0 be the number of pre-intervention periods, $1 \leq T_0 < T$, and let Y_{it}^I be the outcome that would be observed for unit i at time t if the unit i is exposed to the treatment from periods $T_0 + 1$ to T.

The treatment effect for unit i at time t can be indicated by the difference between the observed outcome in the treated unit and the outcome that would have been observed in the absence of the intervention, that is,

$$\alpha_{it} = Y_{it}^I - Y_{it}^N \ , \tag{4}$$

as seen in Equation 2. Let D_{it} be a binary variable that takes the value of 1 if unit i is exposed to the treatment at time t and 0 otherwise, the observed value for unit i at time t is denoted by

$$Y_{it} = Y_{it}^N + \alpha_{it} D_{it} \ ,$$

where Y_{it}^N is not observed (Abadie et al., 2010).

Assuming that the first unit is exposed to the intervention after time T_0, then

$$D_{it} = \begin{cases} 1, & \text{if } i = 1 \text{ and } t > T_0, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Thus, the goal of the synthetic control is to estimate α_{1T_0+1}, α_{1T_0+2}, ..., α_{1T}, and for $t > T_0$,

$$\alpha_{1t} = Y_{1t}^I - Y_{1t}^N = Y_{1t} - Y_{i1}^N \quad .$$

According to Abadie et al. (2010), since Y_{1t} is observed, to estimate the intervention effect, it is necessary to estimate Y_{i1}^N. Suppose Y_{i1}^N is given by a factor model

$$Y_{i1}^N = \delta_t + \theta_t Z_i + \lambda_t \mu_i + \epsilon_{it}, \tag{6}$$

where δ_t is an unknown common factor with constant factor loadings across units, Z_i is a $(r \times 1)$ vector of observed covariates (not affected by the intervention), θ_t is a $(1 \times r)$ vector of unknown parameters, λ_t is a $(1 \times F)$ vector of unobserved common factors, μ_i is a $(F \times 1)$ vector of unknown factor loadings, and ϵ_{it} are the unobserved errors at the region level with zero mean.

Consider a weight vector $\mathbf{W}J \times 1 = (w_2, ..., wJ+1)'$, where $w_j \geq 0$ and $\sum_{j=2}^{J+1} w_j = 1$, then each value w_j represents a potential synthetic control, i.e., the weights of the weighted average of control regions (Abadie et al., 2010).

The value of the response variable for each synthetic control indexed by $\mathbf{W}$ is

$$\sum_{j=2}^{J+1} w_j Y_{jt} = \delta_t + \theta_t \sum_{j=2}^{J+1} w_j \mathbf{Z_j} + \lambda_t \sum_{j=2}^{J+1} w_j \mu_j + \sum_{j=2}^{J+1} w_j \epsilon_{jt} \quad . \tag{7}$$

Suppose there exist $(w_2^{,}..., w_{J+1}^{)}$ such that

$$\sum_{j=2}^{J+1} w_j^* Y_{j1} = Y_{11} \qquad \sum_{j=2}^{J+1} w_j^* Y_{j2} = Y_{12} \qquad ... \qquad \sum_{j=2}^{J+1} w_j^* Y_{jT_0} = Y_{1T_0}$$
$$\sum_{j=2}^{J+1} w_j^* \mathbf{Z_j} = \mathbf{Z_1} \tag{8}$$

Therefore, the estimation of the intervention effect is given by

$$\widehat{\alpha_{1t}} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \tag{9}$$

for $t \in T_0 + 1, ..., T$, $\widehat{\alpha_{1t}}$ is an estimator of α_{1t}. Abadie et al. (2010) point out that it is common for no set of weights to exactly satisfy Equation 8, so the synthetic control region is selected so that Equation 8 is approximately satisfied by the data.

Equation 7 generalizes the usual differences-in-differences model, which can be obtained if λ_t is constant over time, i.e., the model accounts for unobserved confounders as long as they are constant over time. Thus, according to Abadie (2021), the synthetic control method demonstrates the contribution of each unit to the counterfactual of interest; its coefficients are sparse and allow for simple interpretation.

Xu (2017) addresses the synthetic control method for the case of multiple treated units and variable treatment periods, improving the efficiency and interpretability of the method. In the case of multiple treated units, τ and ς represent the group of treated and control units, respectively, and the total number of units is given by $N = N_\tau + N_\varsigma$ (Xu, 2017).

As in the single treated unit model of Abadie et al. (2010), we consider that Y_{it} is the response variable of unit i at time t, and all units are observed over T periods. Let $T_{0,i}$ be the number of pre-treatment periods for unit i, which is first exposed to the treatment at time $(T_{0,i} + 1)$ and subsequently observed for $q_i = T - T_{0,i}$ periods (Xu, 2017). Therefore, with this approach, we can use multiple treated units in different T periods. In the generalized synthetic control (GSC) model proposed by (Xu, 2017), Equation 6 can be rewritten as

$$Y_{it} = \delta_{it} D_{it} + Z_{it}^{'} \beta + \lambda_i^{'} f_t + \epsilon_{it}, \tag{10}$$

where δ_{it} is the heterogeneous treatment effect for unit i at time t, D_{it} is a binary variable equal to 1 if unit i was exposed to the treatment before time t and 0 otherwise, Z_{it} is a $(r \times 1)$ vector of observed covariates (not affected by the intervention), β is a $(1 \times r)$ vector of unknown parameters, λ_i is a $(F \times 1)$ vector of unknown common factors, f_t is a $(F \times 1)$ vector of unobserved common factors, and ϵ_{it} are the unobserved errors for unit i at time t with zero mean.

The GSC estimator for the treatment effect on treated unit i at time t is given by the difference between the actual outcome and its estimated counterfactual, as in Equation 9, with the only difference being that there are multiple treated units. The GSC estimation process involves estimating an interactive fixed-effects model using only the control group and obtaining the coefficients of Equation 10, then estimating the factor loadings for each treated unit by minimizing the mean squared error of the predicted treated outcome in the pre-treatment periods, and finally calculating the treated counterfactuals. More details and explanations about the estimation process are found in Xu (2017).

An estimator for ATT_t is given by

$$\widehat{ATT}_t = \frac{1}{N_\tau} \sum_{i \in \tau} [Y_{it}(1) - \hat{Y}_{it}(0)] \ , \tag{11}$$

for $t > T_0$ (Xu, 2017).

The conditions under which synthetic controls are suitable tools for policy evaluation are discussed by Abadie (2021). The first condition is the Stable Unit Treatment Value Assumption (SUTVA) (Rosenbaum and Rubin, 1983). This assumption implies that there is no interference between units, meaning the outcomes of units are unaffected by the treatments applied to other units. SUTVA ensures the plausibility of causal inference results, ensuring that the treatment does not indirectly interfere with the donor (untreated) units.

Abadie (2021) also highlights the size of the treatment effect and the volatility of the outcome variable, noting that depending on the size of the effect and volatility, it may be difficult to detect in the outcome variable. In addition to this condition, the availability of a comparison group and the absence of anticipation effects are important aspects of synthetic control. The comparison group should represent an adequate group for model inference. To ensure there are available units for the donor group, it is important that not all units adopt interventions similar to the one being investigated during the study period. Finally, it is expected that there is no anticipation of the treatment effects. It is necessary to adjust the data to a period prior to any anticipated effects to estimate the full extent of the intervention's impact (Abadie, 2021).

The biggest challenge in evaluating any treatment is obtaining a reliable estimate of the counterfactual, as it is impossible to observe what would have happened without the treatment. However, to address this problem, the counterfactual outcome can be estimated based on a group of non-participants, and the intervention's impact can be calculated as the difference in the average outcomes between the groups. According to Heinrich et al. (2010), this approach relies on the equivalence between the groups of participants and non-participants in the treatment, with the primary concern being the identification of an appropriate comparison group. In this regard, a mechanism suggested by Abadie (2021) is to restrict the donor group to units with predictor values most similar to the predictor values of the unit exposed to the treatment.

4 Design and data

The bibliometric data from the SCOPUS database is used to track the geographical location of Brazilian researchers in the health and life sciences field, their publications, the "quality" of their published articles, as well as the citations they have received. The unique author identifiers (Scopus author IDs) and their affiliations will be used as the basis for identifying researchers based in Brazil. There is evidence that Scopus's author identification system is reliable for analyzing the migration and mobility of scientists (Aman, 2018), as almost all IDs correctly identify a researcher (Kawashima and Tomizawa, 2015), with 98.4% precision and 90.6% recall (Paturi and Loktev, 2020). Additionally, the

Scopus database has extensive coverage of disciplines such as Natural Sciences, Medicine, Health Sciences, Engineering, and Technology (Pranckutė, 2021) and contains a larger number of Latin American journals in its core collection compared to other databases (Celeste et al., 2021).

The data covers the period from 2000 to 2020, assuming that the country of institutional affiliation of the authors is their geographical location in the year the article is published. The data was extracted from SCOPUS for authors with some Brazilian affiliation during the highlighted period, and then all publications of these researchers were collected. As a quality indicator, the SJR (Scimago Journal Rank) index is used, representing a measure of the prestige of academic journals (McManus et al., 2020).

The raw data is divided into several files, with each file containing all publications per author (author ID). The number of files to be read and organized exceeds 400,000 CSV files, requiring the use of *arrow* (Richardson et al., 2024), *readr* (Wickham et al., 2024), and *purrr* (Wickham and Henry, 2023) packages. After reading and concatenating the data into a single dataframe, we classify the authors into mobility groups, as done in Subbotin and Aref (2021).

The sample of authors is built into six groups of researchers: (i) single article researcher (author of a single publication); (ii) non-migrant researcher (no evidence of international mobility); (iii) immigrant researcher (initial affiliation: other countries, final affiliation: Brazil); (iv) emigrant researcher (initial affiliation: Brazil, final affiliation: other countries); (v) returning migrant researcher (initial affiliation: Brazil, transitional affiliation: other countries, final affiliation: Brazil); (vi) transient migrant researcher (initial affiliation: other countries, transitional affiliation: Brazil, final affiliation: other countries).

The groups used in the impact analyses of quality are those Brazilians who did not return to the country (group iv) and those Brazilians who had an international experience and then returned to Brazil (group v). After classifying the authors according to mobility, the citations are standardized by year of publication, to standardize the number of citations by time of publication. The number of citations per article is winsorized at the 99th percentile. The types of publications used refer to articles and reviews.

The sample used in the impact analysis of mobility consists of emigrant authors and returning migrants who changed countries between 2010 and 2016. The selected treatment period covers a time of significant investments by the Brazilian government in internationalization policies and the expansion of international mobility scholarships in higher education, such as the "Ciencias Sem Fronteiras" (Science without Borders) program (in effect from 2011 to 2016) (Finardi and Guimarães, 2017; Maués and dos Santos Bastos, 2017).

The final data consists of the publications of Brazilian health and life sciences researchers from 2000 to 2020, classified according to the type of mobility. The treated units refer to emigrant and return migrant researchers in multiple treatment periods (2010 to 2016) and non-migrant researchers are the donor units for the synthetic control. Therefore, the donor units are non-mobile scientists, who are those who have never published an article with a "non-Brazilian" affiliation.

Table 1 presents an example of how the database appears after organizing the data with some variables. Additionally, the process of cleaning and organizing the data is available at https://github.com/leobiazoli/thesis.

Table 1: Example of fictitious authorship records from the SCOPUS database

Author ID	DOI	EID	Title	...	Country	Year	Citation	SJR
x	111	2-s2.0-848	...	...	Brazil	2000	10	2.3
x	222	2-s2.0-777	...	...	Brazil	2005	5	3.5
x	333	2-s2.0-555	...	...	Argentina	2020	40	10.7

4.1 Bibliometric mobility measures

Academic mobility is generally identified for authors with at least two publications in the reference period and changes in affiliation and publication sequence (Subbotin and Aref, 2021; Aykac, 2021). In this context, emigrants are defined as those who do not return to their country of origin (Brazil) during the reference period, and returning migrants are defined as those who return to the country where they first declared an affiliation in that country. Bibliometric mobility measures have some limitations, such as issues with author names and multiple affiliations (Appelt et al., 2015). To avoid potential issues, we used the Scopus ID as the identifier for each author and removed multiple affiliations (considering only the first main affiliation).

The number of citations received per article can represent scientific recognition (Reyes-Gonzalez et al., 2016), but it should be considered within the same scientific area for comparative analyses. In turn, the SJR is a bibliometric indicator capable of measuring the prestige or influence of a scientific journal, based on information obtained from the Scopus database (González-Pereira et al., 2010). Additionally, the SJR indicator is a measure that is frequently used to represent the "quality" of scientific publications (Gomez et al., 2020; Horta et al., 2020; Marini and Xu, 2023). Therefore, the number of citations and the SJR are bibliometric measures used in research evaluation analysis systems.

Collaboration in research is also a variable frequently used in bibliometric analyses to study impacts on scientific research (Appelt et al., 2015; Marmolejo-Leyva et al., 2015; Lancho-Barrantes and Cantú-Ortiz, 2019). A proxy for research collaboration used in these studies is co-authorship in scientific publications (Kumar, 2015), which is also utilized here in this database. The collaboration variable is a binary variable that takes the value of 1 when there is at least one co-author with an international (non-Brazilian) affiliation and receives 0 when all authors are Brazilian. Thus, the three bibliometric measures discussed here (citations, SJR, and collaborations) will be analyzed following the mobility of researchers. Table 2 shows the mean and standard deviation for these variables by mobility group.

Table 2: Average and standard deviation of bibliometric data by mobility type

	Citations	SJR index	Collaboration	N. articles	Authors
Single-paper	1.76 ± 2.99	0.64 ± 0.63	0.15 ± 0.36	104,206	122,966
Non-migrant	2.21 ± 3.24	0.78 ± 0.69	0.17 ± 0.37	613,938	182,154
Immigrant	3.37 ± 4.79	1.22 ± 1.15	0.55 ± 0.50	101,335	4,666
Emigrant	4.27 ± 5.69	1.49 ± 1.35	0.73 ± 0.45	123,605	7,408
Return migrant	2.98 ± 4.26	1.09 ± 1.01	0.36 ± 0.48	391,358	15,744
Transient migrant	5.04 ± 6.74	1.76 ± 1.53	0.99 ± 0.12	299,390	4,377

If we observe Table 2, it is evident that mobile scientists (as emigrants and returning migrants) publish in journals of higher prestige and receive more citations than non-mobile scientists (non-migrants). On average, emigrants have almost double the citations and SJR compared to non-migrant authors. In terms of research collaboration, it is also clear that emigrants are able to obtain international co-authors more easily than non-mobile researchers. The number of articles published shows that emigrants publish, on average, 17 articles over the analyzed period, while non-migrants publish approximately 3 articles.

4.2 Impact assessment using synthetic control

The synthetic control method, in addition to accounting for fixed effects, extends the difference-in-differences approach by considering that the unobservable effects of the response variable may vary over time (Abadie et al., 2010). It uses a weighted average of control donors to estimate the causal effects of the treatment, allowing a combination of units in the donor pool to approximate the characteristics of the affected unit substantially better than any unaffected unit alone.

As a recent development of the synthetic control method, a generalization of the synthetic control with multiple treated units has been developed to estimate the average treatment effect (Xu, 2017). Moreover, this generalization of synthetic control can estimate causal effects over multiple treatment periods. However, examples in the literature typically use synthetic control with a single treated unit and at specific points in time, as seen in Farys and Wolbring (2017) and Aykac (2021).

To construct the control group, non-mobile authors with characteristics similar to mobile authors (emigrants and returning migrants) before mobility are used. A vector of covariates $\mathbf{Z}$ is also employed to minimize the discrepancy between the mobile group and the synthetic group before mobility. The covariates used to minimize differences between the treated and control groups include research collaboration, SJR index, and the number of citations.

The generalized synthetic control method is applied to the group of emigrant authors, using the average values of citations, SJR index, and collaborations, across multiple treatment periods (mobility) from 2010 to 2016. In selecting control group donors, the propensity score matching (PSM) method is used to select non-mobile authors with the most similar characteristics to the treated group, based on citations, collaborations, SJR index, and lifetime citations. The Propensity Score Matching (PSM) method is used to correct sample selection bias due to observable differences between the treatment and control groups (Dehejia and Wahba, 2002).

After finding the matched units in the PSM, synthetic control is used with a reduced sample to improve the generation of the counterfactual. Having constructed the corresponding control group of synthetic non-mobile scientists with the closest characteristics to the mobile ones, the synthetic control procedure is followed to estimate the causal effects of scientists' mobility on the number of citations, collaboration patterns, and scientific recognition of publications. Similarly, analyses are conducted for the returning migrant group.

Table 3 shows the averages of the predictor factors before each author's mobility. It can be observed that the averages of the variables presented for the mobile author group (emigrants) and non-mobile author group (non-migrants) show larger differences before the sample selection made by PSM. Therefore, the selection of the most similar authors by PSM succeeded in aligning the variables between the authors, also accounting for the sample reduction. The units treated are the mobile authors with international mobility between 2010 and 2016, which is why the number of mobile authors for both emigrants and return migrants differs from Table 2. Therefore, the donor authors for the synthetic control are constructed to be as close as possible to the mobile authors in terms of these variables. The same situation observed in Table 3 occurs for mobile authors in the group of returned migrants when selecting donors using the PSM. The methodology of this study can be applied to any area of knowledge and any country of origin and destination.

Table 3: Mobile Scientist Means & Predictor Means before the Mobility

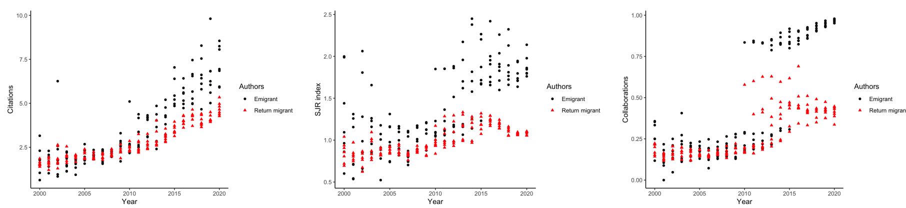
	Actual authors		Sample PSM
	Mobile authors	Non-mobile authors	Non-mobile authors
Citations	2.04	1.49	1.86
SJR index	0.91	0.60	0.81
Collaboration	0.47	0.24	0.41
N. authors	1,453	70,185	3,100

5 Results and discussion

The main descriptive results of the effects of academic mobility on mobile researchers (emigrants and returning migrants) are presented in Figure 1. From the average values of the predictor variables, it is possible to observe the growth in the number of citations, SJR, and international collaborations after the start of mobility events in 2010. The three variables presented in Figure 1 show a more significant evolution for the authors who left the country and did not return compared to those who returned. Additionally, it is possible to observe that the average of the variables before the mobility period is lower than in the post-mobility period. It is important to note that the authors analyzed here refer to those with international mobility between 2010 and 2016.

These results align with the analyses of Franzoni et al. (2014) and Jonkers and Cruz-Castro (2013), which indicate greater productivity and academic prestige for migrant researchers. The higher citation results for emigrants are also observed in Subbotin and Aref (2021), where emigrant researchers have a higher average number of citations than non-migrants. Aykac (2021) also indicates that mobile researchers who remain in a different country from their country of origin benefit more than returning migrant researchers.

Fig. 1: Averages of predictor variables by mobility group



(a) Citations (b) SJR (c) Collaborations

After the descriptive investigations of the differences among mobile researchers, a synthetic trajectory was generated to estimate the effects of academic mobility. The main results on the effects of researcher mobility are represented in Figures 2 and 3. The post-mobilization scientific production of mobile and synthetic scientists is compared in terms of the citations they receive per article, the SJR of the journals in which they are published, and the fraction of their internationally collaborated articles, respectively, in parts a/b, c/d, and e/f.

The increase in citations for articles authored by mobile emigrant scientists can be seen in part a of Figure 2. These researchers manage to receive, on average, 2 times more citations after mobility, with the jump going from 2.21 (sd = 0.30) to 4.55 (sd = 1.32) annual citations for emigrants. Next to Figure 2 is the graph showing the difference between the real and synthetic trajectories of emigrant researchers, along with the 95% confidence interval. Through this graph (Figure 2b) and Table 4, it is possible to observe the values of the Average Treatment Effect on the Treated (ATT) and the differences that are statistically significant at the 5% level.

Following the same logic, the ability to publish in more prestigious academic journals also appears to change after the mobilization of mobile emigrant researchers (Figure 2c), increasing from an average of 0.95 (sd = 0.05) to 1.51 (sd = 0.20). Through the synthetic trajectory (blue dashed line), we can observe the differences in the variables between real and synthetic authors. However, the effects found are not large for all periods and not significant at all treatment points, as shown in Table 4.

The causal effects of mobility on emigrants found in the advances in citations received and the SJR indicator may reveal that international mobility is one of the mechanisms for gaining access to expanded professional networks and knowledge resources (Jonkers and Cruz-Castro, 2013). These findings also align with the research of Marini and Xu (2023), which pointed out that foreign academics' publications attract more citations, tend to be published in higher-ranked journals by Scimago, and also have a higher level of internationalization in terms of co-authorship. Therefore, the results of our research reflect that Brazilian researchers (emigrants) in the health and life sciences field with overseas work experience tend to receive more citations and publish in journals with better bibliometric indicators.

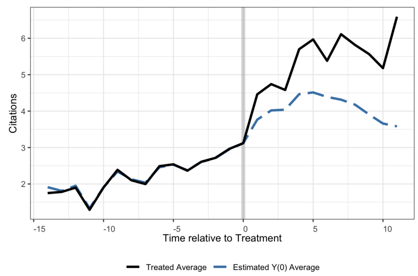
Furthermore, access to international collaboration networks is another point highlighted by Jonkers and Tijssen (2008) that distinguishes mobile authors. Overseas mobility has an important effect on international collaborations. In Figure 2e, collaborations show the proportion of articles in which mobile scientists collaborated with at least one foreign scientist. It is evident in the graph that the international collaboration of mobile authors increased significantly compared to their synthetic counterparts. The sudden increase in international research collaboration is represented in Figure 2e. This variable increases more than fourfold compared to what synthetic scientists achieve in period 0 and stabilizes around an average of 3.06 (sd = 1.61) in subsequent articles, whereas it had an average of 0.63 (sd = 0.09) in international collaboration prior.

The results for returning migrant mobile researchers are presented in Figure 3. However, these findings do not show significant differences, as were found for emigrant authors. It is worth noting, though, that the variables of citations and foreign collaborations had greater changes after international mobility. The gap graphs between the real and synthetic trajectories in Figure 3b and e show consistency in the results obtained for these two variables. The differences found between researchers who do not return to the country (emigrants) and those who do (returning migrants) were also observed in Aykac (2021); Subbotin and Aref (2021).

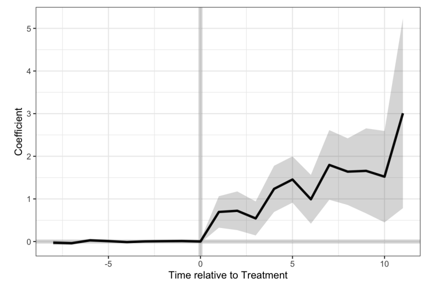
Table 4 shows the estimates of the average treatment effect on the treated (ATT) for the three variables analyzed for emigrant mobile researchers. The second column of this table points to the average effects for citations received and the standard error, then the effects on SJR, and finally, the effects found for international collaborations. Thus, the results found in this study for both emigrants and returning migrants represent gains in access and learning in scientific human capital, as these mobile scientists gain access to resources and also enhance their research capabilities (Ponomariov and Boardman, 2010).

Although previous studies have observed the importance of international scientific mobility (Jonkers and Cruz-Castro, 2013; Franzoni et al., 2014; Robinson-Garcia et al., 2019; Subbotin and Aref, 2021), the results of this research reflect not only the statistical differences found between academic mobility and scien-

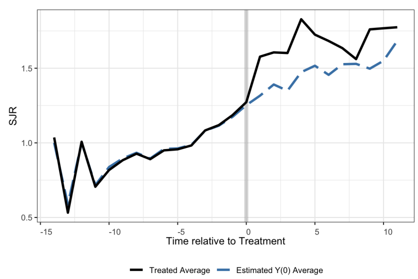
Fig. 2: Synthetic control for Citations, SJR and Collaborations - Emigrants



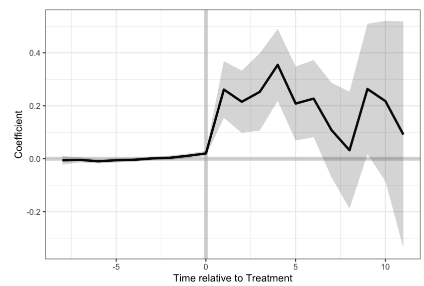
(a)



(b)



(c)



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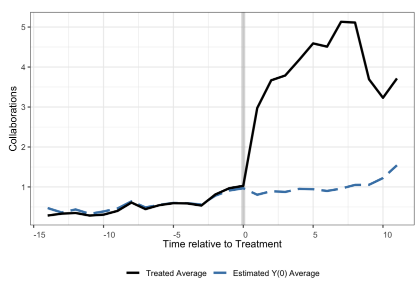

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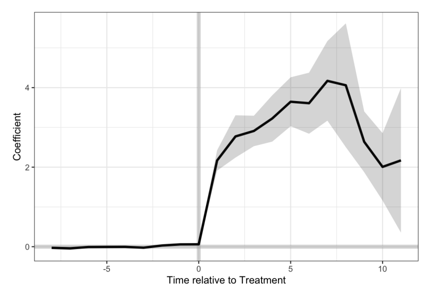


(f)

tific production but also suggest causality between them. Therefore, the results of the synthetic control indicate that international mobility significantly increases the scientific output of mobile authors, especially those who left Brazil and did not return. Thus, the synthetic trajectories from this approach suggest that there is some relationship between the performance of mobile researchers and international mobility.

Emigrants often continue to benefit from access to well-funded research facilities, advanced equipment, and financial resources that are commonly available in developed countries (Franzoni et al., 2014). This access can enhance their research productivity, allowing them to publish more frequently and in higher-impact journals. By contrast, return migrants may lose this direct access upon returning to their home countries, potentially limiting their ability to maintain the same level of research productivity and impact. Limited resources in the home country can inhibit the ability of returnees to undertake cutting-edge research, impacting both the quantity and quality of their publications.

Researchers who remain abroad are more likely to integrate deeply into the prestigious scientific communities of their host countries. This embeddedness allows them to participate in high-impact collaborations and access valuable professional networks that increase their visibility and the likelihood of citations (Jonkers and Tijssen, 2008). Return migrants, however, may face difficulties reintegrating into their home scientific communities, especially if they have been absent for an extended period or if local institutions do not prioritize their international experience (Ackers, 2008). This "brain circulation" dynamic benefits both host and home countries but favors those who stay abroad regarding professional visibility and collaboration opportunities.

The extent and stability of collaborative networks differ significantly between emigrants and return migrants. Emigrants have the advantage of maintaining strong, stable, and ongoing collaborations with their host country colleagues, which facilitates international co-authorship and enhances their research's impact. In contrast, return migrants may experience disruptions in these collaborations upon returning, limiting their opportunities for impactful, international co-authorship. International collaboration is a crucial factor for research visibility and citations, often facilitated by remaining in the host country (Scellato et al., 2015).

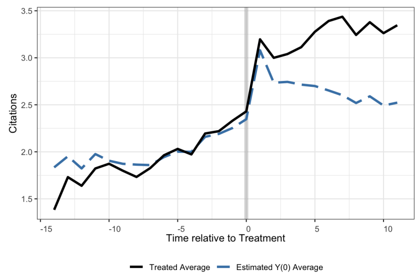
Emigrants are often more attuned to global research trends and priorities, aligning their work with internationally recognized topics that attract higher visibility and citations. This alignment is frequently due to the research agendas of leading global institutions in their host countries, which prioritize research with a broad international appeal. Return migrants, however, may need to adapt their research focus to address local or national needs upon returning, potentially limiting the global relevance of their work.

6 Conclusions

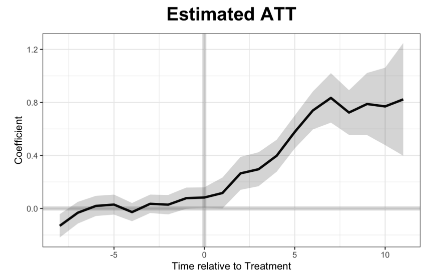
This study was designed to analyze the causal effects of international scientific mobility on the international scientific production of Brazilian health and life sciences researchers, comparing the outcomes of mobile researchers (emigrants and return migrants) with their synthetic counterparts. Existing studies in the literature have already indicated the gains in scientific production generated by academic mobility, as well as its effects on research collaboration networks. However, few studies have employed the synthetic control method to infer causal effects of international mobility on bibliometric variables such as citations, impact factor, and collaborations, especially for Brazilian researchers.

The findings presented here aim to contribute to advances in the studies of international mobility in Brazilian research, regarding recognition, prestige,

Fig. 3: Synthetic control for Citations, SJR and Collaborations - Return migrant



(a)



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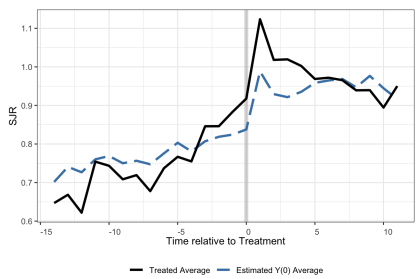

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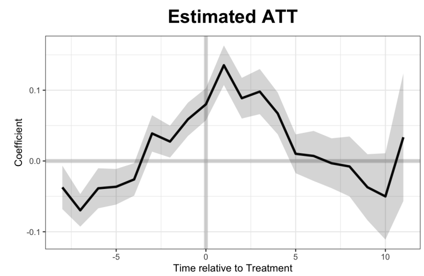

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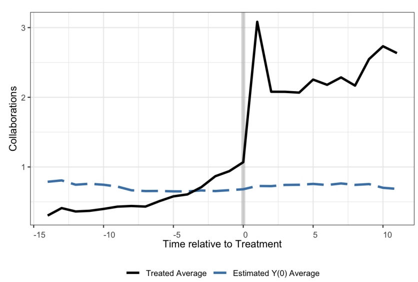

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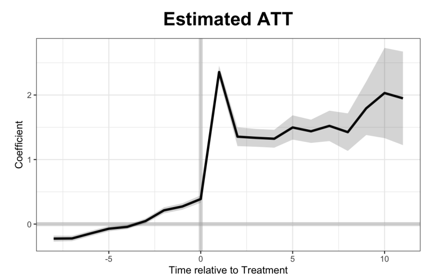

(f)

and the creation of collaboration networks. The synthetic control method was used with multiple treatment periods for mobile researchers who left the country between 2010 and 2016, with the variables of citation count, SJR index, and international collaborations. Analyses were conducted for two groups of researchers, emigrants and return migrants, using the generalization of the synthetic control method and a reduced donor sample. Emigrant mobile scientists receive more citations, publish in more prestigious journals, and collaborate internationally more than their respective counterfactuals created by the synthetic control method.

Table 4: Average treatment effect on the treated (ATT) for emigrants

	$\widehat{ATT}_{Cit}$	$\widehat{ATT}_{SJR}$	$\widehat{ATT}_{Col}$
1	0.69 (0.19)	0.26 (0.05)	2.17 (0.13)
2	0.72 (0.23)	0.22 (0.06)	2.77 (0.27)
3	0.54 (0.20)	0.25 (0.07)	2.91 (0.20)
4	1.24 (0.28)	0.35 (0.07)	3.23 (0.30)
5	1.46 (0.28)	0.21 (0.07)	3.65 (0.31)
6	0.99 (0.29)	0.23 (0.07)	3.61 (0.39)
7	1.80 (0.42)	0.11 (0.09)	4.17 (0.51)
8	1.64 (0.40)	0.03 (0.11)	4.06 (0.79)
9	1.66 (0.51)	0.26 (0.13)	2.64 (0.39)
10	1.52 (0.55)	0.22 (0.15)	2.01 (0.43)
11	3.01 (1.14)	0.09 (0.22)	2.17 (0.93)

The results for researchers who migrated and then returned to Brazil indicate that they do not benefit as much as emigrants from international mobility, but there is still a significant increase in international collaborations after leaving the country. Therefore, this study highlights the importance of international mobility for Brazilian science, both at the individual and collective levels, revealing its magnitude as well as the causal relationship between them. However, several limitations of this study must be considered.

The main limitation refers to the exclusive use of bibliometric data from Scopus, which may prevent the inclusion of confounding variables that bring changes to the variables studied, such as salaries, researchers' involvement in non-publication work, teaching skills, etc. Moreover, a point not addressed was the use of controls such as gender, host country, and field of study (subfield), to assess how the effects might differ. An additional study could usefully explore how these controls play a role in the effects of international mobility.

Finally, it is worth noting that this study suggests several directions for the Brazilian government. It highlights strategies the government should consider in its public policies for the internationalization of higher education, aiming to plan exchanges that foster national scientific development. Another topic that also merits discussion is the attempt to recover mobile researchers who do not return to Brazil, a subject currently being addressed by the Brazilian federal government through the "Programa de Repatriação de Talentos - Conhecimento Brasil" (Talent Repatriation Program - Knowledge Brazil). This federal policy aims to attract and retain qualified human resources in the country, beginning in 2024. It would also be beneficial for the development of national science to motivate local researchers to gain more scientifically impactful experience abroad (including scholarships, grants, or awards) that specifically benefits the most productive scientists upon their return.

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Authors' Contributions

Leonardo Biazoli: Conceptualization, Methodology, Formal analysis, Data Curation and Writing - Original Draft.
Bruna de Paula Fonseca: Conceptualization, Methodology and Writing - Review & Editing.
Mine Çetinkaya-Rundel: Conceptualization, Methodology and Data analysis - Review & Editing.
Eric Fernandes de Mello Araújo: Conceptualization, Methodology, Data collection and Writing - Review & Editing.
Izabela Regina Cardoso de Oliveira: Conceptualization, Methodology and Writing - Review & Editing.

Statements and Declarations

The authors have no competing interests.

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Data availability

The codes used in this work, including pre-processing, organization and cleaning, descriptive analysis and synthetic control analysis, are publicly available on *Github* at https://github.com/leobiazoli/thesis.

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capesData: Data on Scholarships in CAPES International Mobility Programs

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Abstract. *capesData* is an R package that provides an interface for accessing data on scholarships for international mobility programs funded by the Brazilian government through CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior). The package provides access to information such as the type of scholarship program, the start and end dates of the scholarship, the home institution, the study institution, the country of study, the program name, and other variables related to the financial aid received by the scholars. The package provides information on scholars for the period between 2010 and 2019 and it contains two datasets. The first dataset contains only 14 variables and 146,036 observations, while the second dataset has 51 variables and the same number of observations. The package automates common data reading and organization steps and allows researchers to easily and transparently track the origin of the underlying data sources. It is designed to enable users to easily extend the package's features and contribute to shared data processing. The source code for the package code is version tracked and hosted on GitHub and CRAN (The Comprehensive R Archive Network).

Keywords: Mobility program; Scholarship; Researcher; Package; Brazil.

1 Statement of need

Evaluations of the effects of brain drain can be conducted from various perspectives and using different statistical models. One of the challenges in assessing international brain migration is the lack of organized and reliable databases over a long observation period. The works of Aref et al. (2019), Subbotin and Aref (2021), Aykac (2021), Kotsemir et al. (2022) and Poitras and Larivière (2023) utilize bibliometric databases, such as Scopus and Web of Science, to study the academic impacts of researchers' migratory movements. Baruffaldi et al. (2020) used multiple data sources to analyze the effect of an international mobility scholarship on research, including government data and data collected from candidates' curricula, LinkedIn profiles, and personal web pages. However, as highlighted by Subbotin and Aref (2021), this type of database (bibliometric data) may have limitations, causing not only random noise but also systematic biases.

An alternative to minimize errors and work with reliable databases is to use government statistics provided by government agencies. At the national level, data on international academic mobility of researchers funded by the Brazilian government are available for download from official sources such as the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) (CAPES, nd) or the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) (CNPq, nd). However, these governmental agencies provide data related to international academic mobility scholarships separated by year and/or programs, and this is generally the only way to access data on a national scale.

Sometimes gathered from multiple databases, for example, separated by year, this data comes in various formats and with different names for the same variables, requiring users to verify and standardize the data before it can be combined or processed for analysis. This creates the possibility of errors through programming mistakes, changes in variable names, or unexpected alterations in a data source. This can lead to misrepresentation of the data in ways that are difficult to identify, such as treating the same variable as two different variables.

Due to these issues, it is important to develop robust tools that provide clean, verified, and standardized data from multiple sources in a transparent manner. The *capesData* package provides easy access to clean data, ready to analyze the data of Brazilian researchers in government-funded scholarship programs for international academic mobility, and in a structure that is easy to track from raw data to the final set of standardized data. Additional arguments for this package allow users, among other options, to easily manipulate and visualize the data directly in R with two datasets: one complete set of raw data and another with summarized variables.

The *capesData* package relies heavily on popular packages that many researchers are familiar with (such as dplyr (Wickham et al., 2023), tidyr (Wickham et al., 2024), and others from the tidyverse (Wickham et al., 2019)), and therefore, it can be easily adopted by researchers working in R. Currently, *capesData* provides data collected by CAPES/Brazil for Brazilian researchers with international mobility scholarships from 2010 to 2019. The *capesData* package was developed to facilitate access, manipulation, and analysis of data from CAPES database. CAPES is a key Brazilian government agency responsible for the evaluation and support of higher education programs. The dataset contains valuable information for researchers, policymakers, and educational institutions.

2 State of the field

Some R packages provide routines and bibliometric data that can be used to perform bibliometric data analyses but can also be utilized to track researcher mobility based on their publications, such as bibliometrix and bibliometrixData (Aria and Cuccurullo, 2017; Aria, 2022). The bibliometrix package provides various routines to import bibliographic data from Scopus, Clarivate Analytics' Web of Science, PubMed, Digital Science Dimensions, and Cochrane databases, perform bibliometric analyses, and create data matrices for co-citation, coupling, scientific collaboration analysis, and co-word analysis. The bibliometrixData package includes example datasets extracted from Scopus and Clarivate Analytics' Web of Science.

Other interfaces that more closely resemble the purpose of the *capesData* package are the getLattes and getLattesData packages (Souza and Sabino, 2020; Perlin, 2024). These packages provide information about researchers registered in the curriculum database for academics across the country, with over 5 million records, known as "Lattes" (¡http://lattes.cnpq.br/¿). The academic life of Brazilian researchers, or those related to Brazilian universities, is documented in Lattes. Some of the information that can be obtained includes professional training, research area, publications, academic advisory roles, projects, etc. The getLattes package allows working with Lattes data exported to XML format. The getLattesData package is a set of functions that I have been using to access the data. Therefore, using this package requires the manual download of zip files with the XML data.

The CAPES is a central Brazilian government agency responsible for overseeing and evaluating the quality of graduate programs and promoting scientific research and advanced studies in Brazil. CAPES maintains a comprehensive database that includes information on faculty, students, courses, publications, and institutional evaluations. This database is a vital resource for educational research, policy analysis, and institutional reporting. The *capesData* package provides a comprehensive and easy-to-use tool specifically designed to address the unique challenges of working with CAPES data. By automating data access and cleaning and offering robust data manipulation and visualization capabilities, *capesData* significantly enhances the efficiency and accessibility of CAPES data analysis. This package represents a significant advancement in empowering researchers, policymakers, and educational institutions to make data-driven decisions based on reliable and timely insights from CAPES data.

3 Future Directions

Educational data analysis is constantly evolving, with increasing trends toward automation, integration, and accessibility. In the future, the field is likely to progress in the following directions:

Data Integration: Improved integration with other educational and research databases to provide a more comprehensive view of educational outcomes.

Advanced Analytics: Incorporation of advanced analytical techniques, including machine learning and artificial intelligence, to uncover deeper insights from educational data.

The *capesData* package is well-positioned to adapt to these trends and continue providing valuable tools and capabilities to its users. By staying at the cutting edge of technology and user needs, *capesData* aims to contribute significantly to the advancement of educational research and policy analysis in Brazil and beyond.

4 Examples of use

The installation of the *capesData* package can be performed either through CRAN, which is the official R repository network, or directly via GitHub, where the most up-to-date version of the package is available. After installing *capesData*, users gain access to two main datasets. The first, *capesData_raw*, includes all variables provided by CAPES, allowing for a more comprehensive and detailed analysis of the information. The second dataset, *capesData*, contains a simplified version focusing on the most relevant variables for analyzing mobility programs.

The information contained in these datasets enables the exploration of various aspects of international mobility programs. For instance, the data reveal details about the number of scholarships awarded by the level of study (Table 1), the temporal distribution of scholarships over the years (Figure 1), and the fields of knowledge most covered by the programs (Figure 2). These results offer a broad and detailed view of the impact and scope of the mobility programs.

To efficiently extract and organize this information, we used a combination of powerful and widely utilized packages in the R environment, such as *tidyverse*, which facilitates data manipulation and visualization, and *dplyr*, which provides intuitive functions for handling tables and aggregating data. To generate well-formatted and professional tables, we used the *kableExtra* package, which allows for highly customizable tables.

For graphical visualizations, we employed the *ggplot2* package, known for its flexibility and high-quality graphics. To generate maps with geographic locations, we added the *sf*, *rnaturalearth*, and *rnaturalearthdata* packages. These packages enable spatial data manipulation and the creation of detailed maps. With this approach, we extracted relevant insights and communicated the results in a clear, structured, and visually impactful manner.

Table 1: Distribution of scholarships by level of study from 2010 to 2019

Level of study	**Frequency**
Graduação Sanduíche	85548
Doutorado Sanduíche	34089
Doutorado Pleno	8653
Estágio Pós-Doutoral	6789
Estágio Sênior	2962
Pós-Doutorado	1535
Pesquisador Visitante Especial	1179
Capacitação Professores Da Educação Básica	783
Mestrado Profissional Nos EUA	560
Professor Visitante	483
Outros	3455

Fig. 1: Mobility grants per year

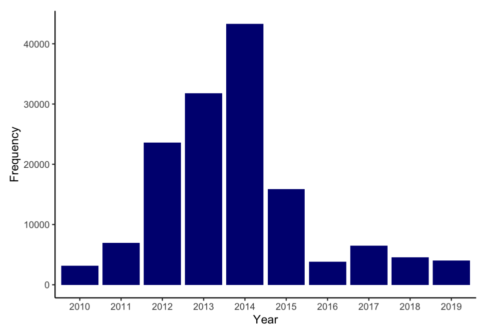


Fig. 2: Mobility grants by area

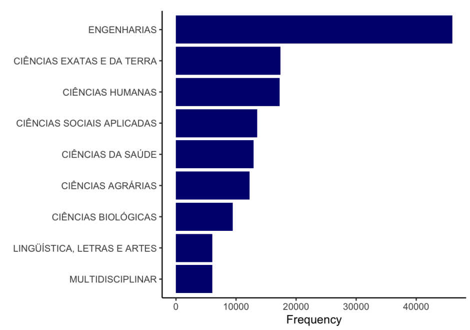
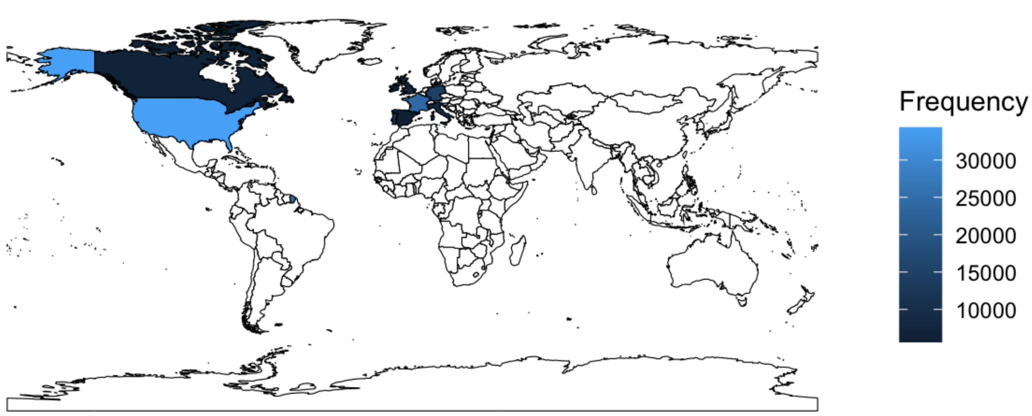


Fig. 3: Mobility grants by country of destination



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Package ‘capesData’

August 28, 2024

Title Data on Scholarships in CAPES International Mobility Programs

Version 0.0.1

Description Information on activities to promote scholarships in Brazil and abroad for international mobility programs, recorded in Capes' computerized payment systems. The CAPES database refers to international mobility programs for the period from 2010 to 2019 <https://dadosabertos.capes.gov.br/dataset/>.

License CC0

Encoding UTF-8

RoxygenNote 7.3.1

Depends R (>= 2.10)

LazyData true

LazyDataCompression xz

NeedsCompilation no

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Repository CRAN

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Contents

capesData *Data on Scholarships in CAPES International Mobility Programs*

Description

The goal of capesData is to provide an attractive dataset for exploring and visualizing data on the International Mobility Programs funded by CAPES/Brazil. The CAPES database refers to international mobility programs for the period from 2010 to 2019.

Usage

```
capesData
```

Format

A data frame with 146036 rows and 14 columns:

ID_PROCESSO Number of concession process.

NM_NIVEL Type of scholarship.

AN_INICIO Year in which the scholarship begins.

ME_INICIO Month in which the scholarship begins.

AN_FIM End year of the scholarship.

ME_FIM End month of the scholarship.

NM_IES_ORIGEM_PRINCIPAL_DA Name of the beneficiary's home institution.

NM_UF_IES_ORIGEM Name of the Federation State of the beneficiary's institution of origin.

NM_IES_ESTUDO_PRINCIPAL_DA Name of the beneficiary's study institution.

NM_PAIS_IES_ESTUDO Country of destination of the beneficiary.

NM_GRANDE_AREA Field of knowledge.

NM_AREA_CONHECIMENTO Area of knowledge classified by CAPES.

NM_AREA_AVALIACAO Area of project evaluation.

NM_PROGRAMA Program Name.

Source

Created by Leonardo Biazoli with data provided by CAPES/BRAZIL https://dadosabertos.capes.gov.br/dataset/.

See Also

capesData_raw

Examples

```
capesData
```

capesData_raw *Data on Scholarships in CAPES International Mobility Programs*

Description

The goal of capesData is to provide an attractive dataset for exploring and visualizing data on the International Mobility Programs funded by CAPES/Brazil. The CAPES database refers to international mobility programs for the period from 2010 to 2019.

Usage

```
capesData_raw
```

Format

A data frame with 146036 rows and 51 columns:

ID_PROCESSO Number of concession process.

NM_NIVEL Type of scholarship.

AN_INICIO Year in which the scholarship begins.

ME_INICIO Month in which the scholarship begins.

AN_FIM End year of the scholarship.

ME_FIM End month of the scholarship.

NM_IES_ORIGEM_PRINCIPAL_DA Name of the beneficiary's home institution.

NM_UF_IES_ORIGEM Name of the Federation State of the beneficiary's institution of origin.

NM_IES_ESTUDO_PRINCIPAL_DA Name of the beneficiary's study institution.

NM_PAIS_IES_ESTUDO Country of destination of the beneficiary.

NM_GRANDE_AREA Field of knowledge.

NM_AREA_CONHECIMENTO Area of knowledge classified by CAPES.

NM_AREA_AVALIACAO Area of project evaluation.

NM_PROGRAMA Program Name.

CD_MOEDA Functional payment currency.

VL_TOTAL_RECEBIDO_MOEDA Total amount received in functional currency.

VL_AUX_DESLOCAMENTO Amount received as travel assistance.

VL_ADICIONAL_LOCALIDADE Amount received as a locality bonus.

VL_ADICIONAL_DEPENDENTE Additional amount paid to full doctoral scholarship holders with registered dependents.

VL_BOLSA Amount received as a scholarship/fees.

VL_TAXA_ESCOLAR Amount received for tuition and fees.

VL_DIF_AUX_DESLOCAMENTO Amount received as differences/reimbursements for travel expenses that were performed.

VL_DIF_SEGURO_SAUDE Amount received as differences/reimbursements for expenses performed with health insurance.

VL_SEGURO_SAUDE Amount received as health insurance.

VL_TAXA_ESCOLAR_VALOR_P Amount received for small tuition and fees.

VL_OUTROS_CREDITOS Amount received for the payment of extraordinary expenses.

VL_DIF_MENSALIDADE Amount received as differences/reimbursements for expenses performed with tuition.

VL_OUTROS_DEBITOS Amount received as differences/refunds for other debts.

VL_DIF_DEPENDENTES Amount received as differences/reimbursements for expenses performed with additional dependent.

VL_MENSALIDADE Amount received as a stipend.

VL_AUX_INSTALACAO Amount received as installation aid.

VL_AUX_SEGURO_SAUDE_ANUAL Amount received in annual installments as health insurance.

VL_MENSALIDADES_AGENDADAS Amount received as scheduled tuition.

VL_REEMBOLSO_TAXA_ESCOLAR Amount received for the refund of tuition and fees paid by the beneficiary.

VL_AUX_MORADIA Amount received as housing allowance.

VL_DIF_AUX_INSTALACAO Amount received as differences/refunds for installation aid.

VL_REEMBOLSO_PASSAGEM_AEREA Amount received as differences/refunds for airfare.

VL_AUX_MATERIAL_DIDATICO Amount received as aid for teaching materials, paid under the Ciencia Sem Fronteiras Program.

VL_AUX_SEGURO_SAUDE Amount received as health insurance benefit.

VL_AJUDA_CUSTO_CAPES_SETEC Amount received as a subsistence allowance, paid under the CAPES/SETEC program.

VL_MENSALIDADE_DEMANDA Amount received as a monthly fee on demand.

VL_AUX_DESLOCAMENTO_DEMANDA Amount received as travel allowance on demand.

VL_MENSALIDADE_LIC_MATERNIDADE Amount received as tuition during maternity leave by the scholarship holder during the duration of the process.

VL_ESCOLA_ALTOS_ESTUDOS Amount received as a scholarship under the Escola de Altos Estudos program.

VL_AJUDA_CUSTO Amount received as a subsistence allowance.

VL_AUX_INSTALACAO_DEPENDENTE Additional amount received as installation aid, paid to full doctoral scholarship holders with registered dependents.

VL_AUX_SEGURO_SAUDE_DEPENDENTE Additional amount received as health insurance aid, paid to full doctoral scholarship holders with registered dependents.

VL_AJUDA_CUSTO_MTUR Amount received as a subsistence allowance, paid under the CAPES/MTUR program.

VL_AUX_DESLOCAMENTO_RETORNO_D Amount received as travel allowance for return.

VL_AUX_SEGURO_SAUDE_DEMANDA Amount received as health insurance on demand.

VL_AUX_DESLOCAMENTO_PESQUISA Amount received as travel allowance for research.

Source

Created by Leonardo Biazoli with data provided by CAPES/BRAZIL https://dadosabertos.capes.gov.br/dataset/.

See Also

capesData

Examples

```
capesData_raw
```

THIRD PART

5 FINAL CONSIDERATIONS

This study was designed to investigate patterns of academic mobility among researchers in the health field, as well as scientific productivity and recognition of their publications, based on bibliometric data from Scopus. The results observed here aim to contribute to the advancement of studies on international academic mobility in Brazilian research, particularly regarding recognition, prestige, and the creation of collaboration networks. In this study, we seek to detail the methodological processes required for data collection, analysis, and visualization, so as to enable the application of these processes in other bibliometric studies across different areas and countries.

Exploratory analyses of researchers in the health sciences and life sciences reveal that migrant researchers, although fewer in number, contribute significantly to the development of national science, with bibliometric indicators showing higher averages compared to non-migrant researchers. Additionally, the main destinations involved in the academic mobility of these health researchers were the United States, Canada, the United Kingdom, and France. Furthermore, researchers in the field of Medicine represented the largest group in the dataset and also the largest number of migrants.

When analyzing the causal effects of international scientific mobility on the international scientific output of Brazilian researchers, comparing the results of mobile researchers (emigrants and return migrants) with their synthetic counterparts, we observed that emigrant mobile scientists receive more citations, publish in more prestigious journals, and collaborate more internationally. The results for researchers who migrated and then returned to Brazil indicate that they do not benefit as much from international mobility as emigrants, but still show significant increases in international collaborations after their departure.

The use of the synthetic control method with multiple treated periods for Brazilian mobile researchers in the health sciences and life sciences was an unprecedented application and also advanced with the pre-selection of the control group through the propensity score matching method. The main limitation relates to the exclusive use of bibliometric data from Scopus, which may prevent the use of confounding variables that introduce changes to the studied variables, such as salaries, researcher involvement in non-publication-based work, teaching skills, etc. Additionally, an issue not addressed was the use of controls such as gender, host country, and field of study to assess how the effects may differ. An additional study could usefully explore how these controls play a role in the effects of international mobility.

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